COMPUTER NETWORK : MODULE 1

Introductory Concepts: Goals and applications of networks, Categories of networks, Organization of the Internet, ISP, Network structure and architecture (layering principles, services, protocols and standards), The OSI reference model, TCP/IP protocol suite, Network devices and components. Physical Layer: Network topology design, Types of connections, Transmission media, Signal transmission and encoding, Network performance and transmission impairments, Switching techniques and multiplexing.

Data Communications

For communication of information and messages we use telephone and postal communication systems.

Data and Information from one computer system can be transmitted to other systems across geographical areas. In these methods it includes electrical signals carried along a conductor, optical signals along optical fibers and electromagnetic areas.

Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Basic Elements of a Data Communication



I. Message. The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.

2 Sender. The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.

3. Receiver. The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.

4. Transmission medium. The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.

5. Protocol. A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices.

Characteristics

Delivery: Data must be delivered to the correct destination and must be received by the intended receiver only.

Accuracy: Data delivered must be accurate. Any alteration to the data during transmission renders it useless.

Timeliness: Data must be delivered within the ideal time else it is useless. In the case of audio and video, the data must be delivered as they are produced; in the same order they are produced without significant delay. This kind of delivery is called real-time transmission.

Jitter: It is the variation in the arrival time of audio or video packets or unevenly delays in the delivery of the audio or video packets.

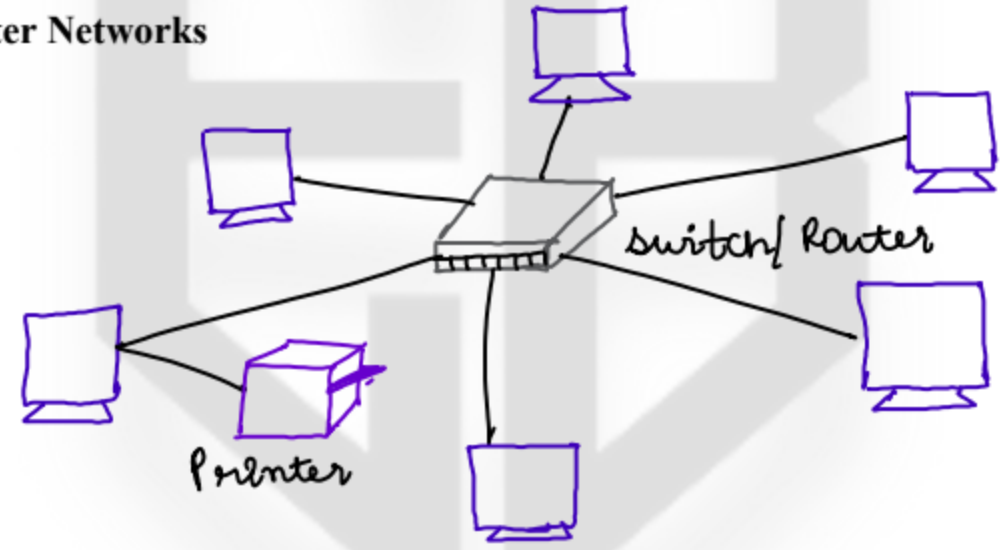
Data Representation

- Text
- Numbers
- Images
- Audio
- Video

Networks

A network is a set of devices (often referred to as nodes) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network. Computer network connects two or more autonomous computers. The computers can be geographically located anywhere.

Computer Networks



A computer network is an interconnection of various computer systems located at different places. Two or more computers are linked together with a medium and data communication devices for the purpose of communicating data and sharing resources. The computer that provides resources to other computers on a network is known as a server. Individual computers in the network, which access shared network resources, are known as nodes.

Uses of Computer Networks

Business Applications: In the context of business applications, CN can refer to various things, such as: Customer Networks: Companies can use CN to connect with their customers and build relationships. For example, social media platforms like Facebook and Twitter enable companies to engage with their customers and gather feedback.

Corporate Networks: Companies can use CN to connect their employees and facilitate communication, collaboration, and resource sharing. For example, companies can use intranets or enterprise social networks to allow employees to access company resources, such as files, databases, and applications.

Cloud Networks: Companies can use CN to access cloud services, such as Software as a Service (SaaS), Infrastructure as a Service (IaaS), or Platform as a Service (PaaS). For example, companies can use cloud-based productivity tools like Google Docs or Office 365 to collaborate on documents.

Entertainment Networks: CN can be used to stream entertainment content, such as movies, music, or games. For example, users can use streaming services like Netflix or Spotify to access a vast library of entertainment content.

Mobile Users: In the context of mobile users, CN can refer to various things, such as: Mobile Networks: CN can be used to connect mobile devices, such as smartphones or tablets, to the internet or other devices. For example, users can use mobile data or Wi-Fi to access the internet or connect to other devices.

Social Issues: In the context of social issues, CN can refer to various things, such as: Social Networks: CN can be used to connect people with similar interests or backgrounds. For example, social networks like Facebook, Twitter, or LinkedIn enable users to connect with friends, family, colleagues, or like-minded people.

Goals and Applications of Networks

Resource sharing: Computer networks enable multiple devices to share resources such as printers, storage devices, and internet connections. This can help reduce costs and improve efficiency.

Communication: Computer networks enable communication between users, whether it's through email, instant messaging, or video conferencing. This can help improve collaboration and productivity.

Data sharing: Computer networks enable users to share data and information, whether it's through file sharing or cloud-based collaboration tools. This can help improve decision-making and innovation.

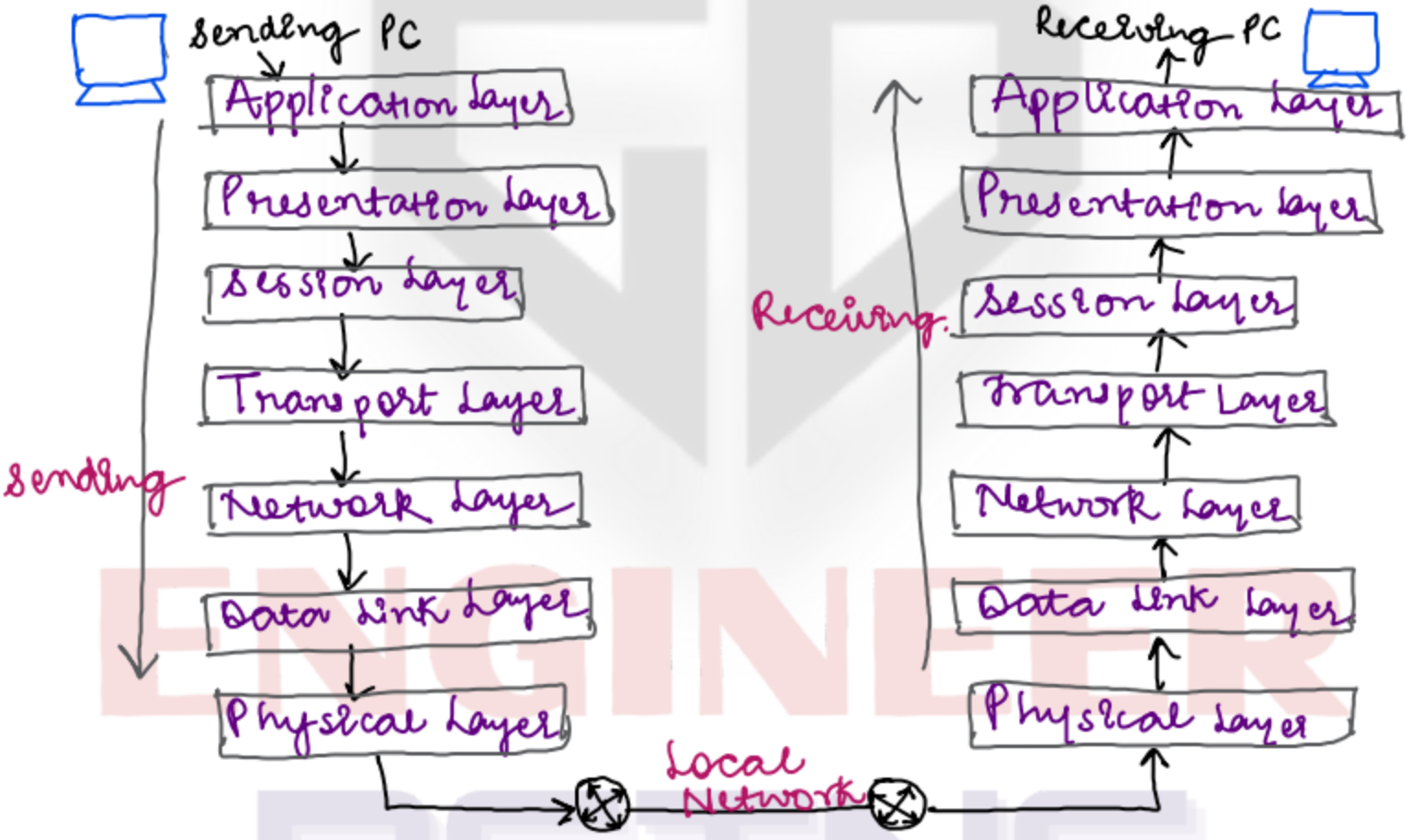
Backup and recovery: Computer networks enable backup and recovery of data and applications, helping to protect against data loss and ensuring business continuity in the event of a disaster.

Internet of Things (IoT): Computer networks enable the IoT, where devices such as sensors, cameras, and other smart devices can be connected to the internet and communicate with each other.

Network Model

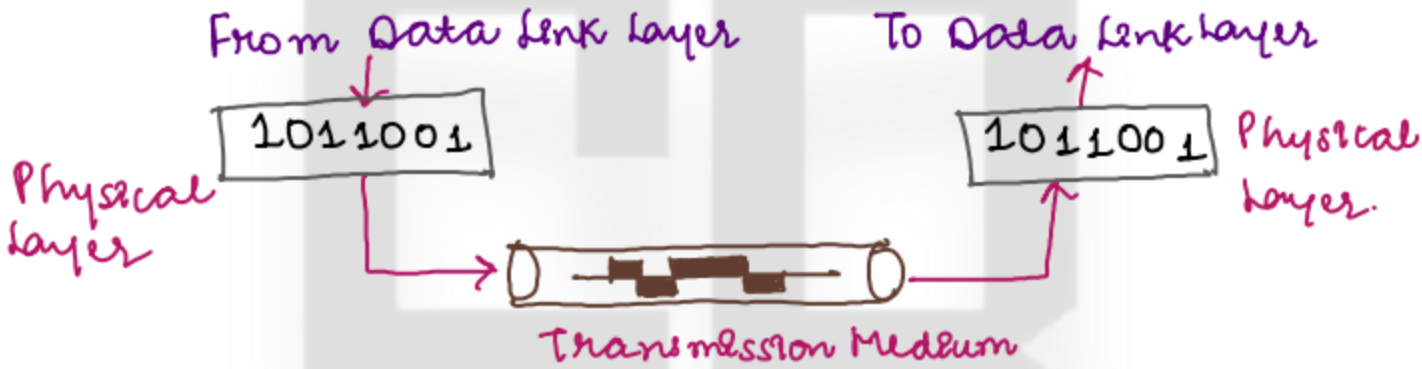
- OSI
- TCP/IP

OSI (Open Systems Interconnection)



The Open Systems Interconnection (OSI) model defines network communication as a series of layers, with each layer responsible for a specific aspect of network communication.

Physical Layer



The lowest layer of the OSI reference model is the physical layer. It is responsible for the actual physical connection between the devices. The physical layer contains information in the form of bits. It is responsible for transmitting individual bits from one node to the next. When receiving data, this layer will get the signal received and convert it into 0s and 1s and send them to the Data Link layer, which will put the frame back together.

The functions of the physical layer are:

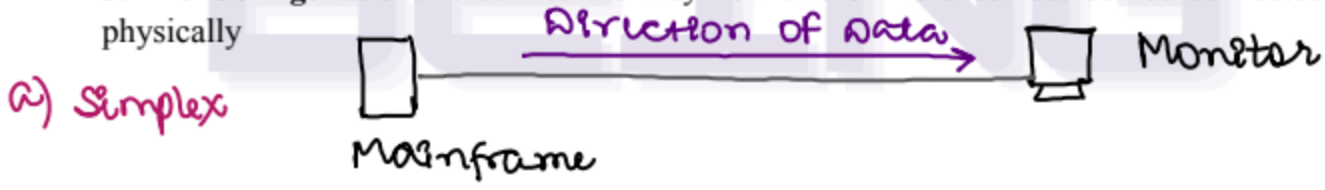
1. Bit synchronization: The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at bit level.

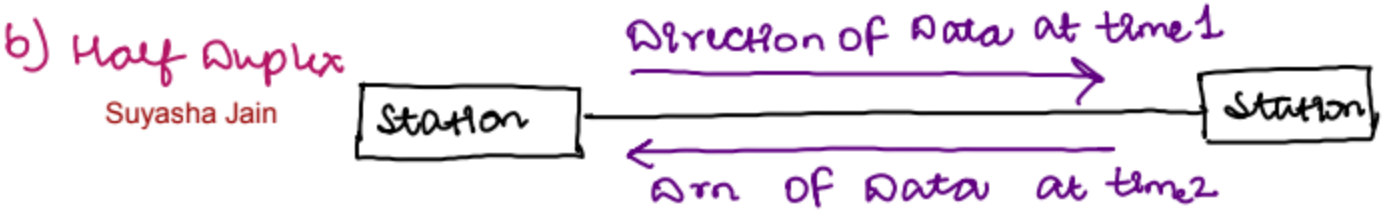
2. Bit rate control: The Physical layer also defines the transmission rate i.e. the number of bits sent per second.

3. Physical topologies: Physical layer specifies the way in which the different devices/nodes are arranged in a network i.e. bus, star, or mesh topology.

4. Transmission mode: Physical layer also defines the way in which the data flows between the two connected devices. The various transmission modes possible are: Simplex, half-duplex and full duplex.

5.Line Configuration: It defines the way how two or more devices can be connected physically



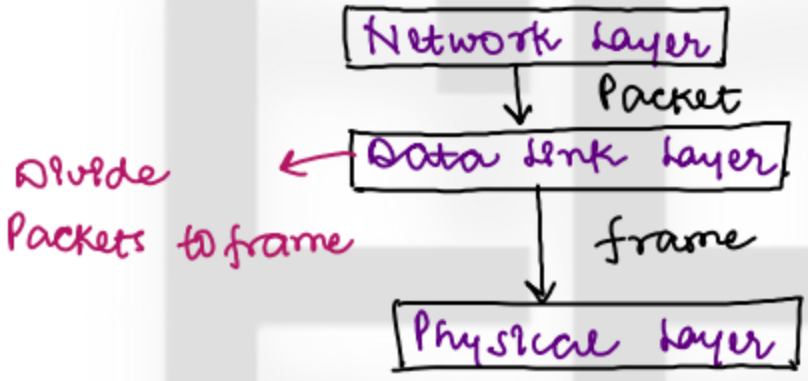


Suyasha Jain



Data Link Layer

The data link layer is responsible for the node to node delivery of the message.



The functions of the data Link layer are :

1. Framing: Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. This can be accomplished by attaching special bit patterns to the beginning and end of the frame.

2. Physical addressing: After creating frames, Data link layer adds physical addresses (MAC address) of sender and/or receiver in the header of each frame.

3. Error control: Data link layer provides the mechanism of error control in which it detects and retransmits damaged or lost frames.

4. Flow Control: The data rate must be constant on both sides else the data may get corrupted thus , flow control coordinates that amount of data that can be sent before receiving acknowledgement.

5. Access control: When a single communication channel is shared by multiple devices, MAC sublayer of data link layer helps to determine which device has control over the channel at a given time.

Network Layer (Layer 3)

Network layer works for the transmission of data from one host to the other located in different networks. It also takes care of packet routing i.e. selection of the shortest path to transmit the packet, from the number of routes available. The sender & receiver's IP address are placed in the header by the network layer.

The functions of the Network layer are

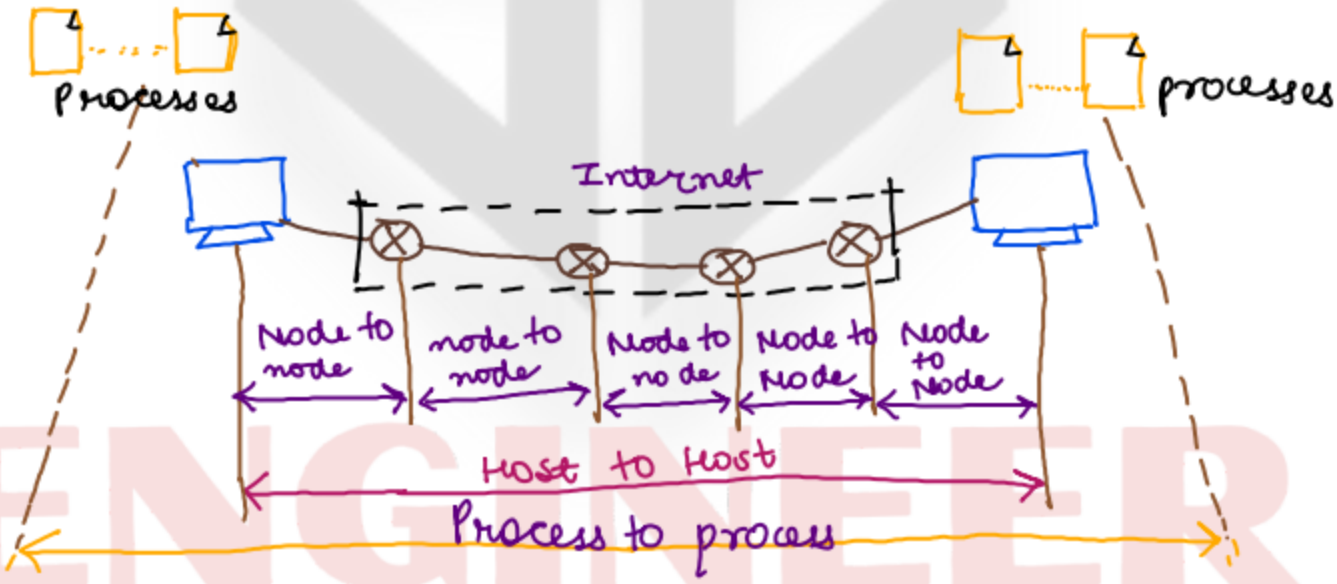
1. Routing: The network layer protocols determine which route is suitable from source to destination. This function of the network layer is known as routing.

2. Logical Addressing: In order to identify each device on internetwork uniquely, the network layer defines an addressing scheme. The sender & receiver's IP address are placed in the header by the network layer. Such an address distinguishes each device uniquely and universally.

* Segment in Network layer is referred as Packet.

** Network layer is implemented by networking devices such as routers

Transport Layer



1.Segmentation and Reassembly: This layer accepts the message from the (session) layer, breaks the message into smaller units. Each of the segments produced has a header associated with it. The transport layer at the destination station reassembles the message.

2. Service Point Addressing: To deliver the message to the correct process, the transport layer header includes a type of address called service point address or port address. Thus, by specifying this address, the transport layer makes sure that the message is delivered to the correct process.

The services provided by the transport layer:

1. Connection Oriented Service: It is a three-phase process which include

– Connection Establishment

– Data Transfer

– Termination / disconnection

In this type of transmission, the receiving device sends an acknowledgement back to the source after a packet or group of packets is received. This type of transmission is reliable and secure.

2. Connection less service: It is a one-phase process and includes Data Transfer. In this type of transmission, the receiver does not acknowledge receipt of a packet. This approach allows for much faster communication between devices. Connection-oriented service is more reliable than connectionless Service.

* Data in the Transport Layer is called Segments.

** Transport layer is operated by the Operating System. It is a part of the OS and communicates with the Application Layer by making system calls. Transport Layer is called as Heart of OSI model

Session Layer

This layer is responsible for establishment of connection, maintenance of sessions, authentication and ensures security.

The functions of the session layer are:

1. Session establishment, maintenance, and termination: The layer allows the two processes to establish, use and terminate a connection.

2. Synchronization: This layer allows a process to add checkpoints which are considered as synchronization points into the data. These synchronization points help to identify the error so that the data is re-synchronized properly and ends of the messages are not cut prematurely and data loss is avoided.

3. Dialog Controller: The session layer allows two systems to start communication with each other in half-duplex or full duplex.

Presentation Layer

Presentation Layer (Layer 6) Presentation layer is also called the Translation layer. The data from the application layer is extracted here and manipulated as per the required format to transmit over the network. The functions of the presentation layer are:

1. Translation: For example, ASCII to EBCDIC(Extended Binary Coded Decimal Interchange Code).

2. Encryption/ Decryption: Data encryption translates the data into another form or code. The encrypted data is known as the cipher text and the decrypted data is known as plain text. A key value is used for encrypting as well as decrypting data.

3. Compression: Reduces the number of bits that need to be transmitted on the network.

Application Layer

Application Layer (Layer 7)The application layer is the topmost layer of the OSI (Open Systems Interconnection) model and is responsible for providing application services to user processes, such as file transfer, email, and web browsing. The role of the application layer is to enable communication between applications running on different networked devices.

**Application Layer is also called Desktop Layer.

The functions of the Application layer are :

1. Network Virtual Terminal

2. FTAM-File transfer access and management

3. Mail Services

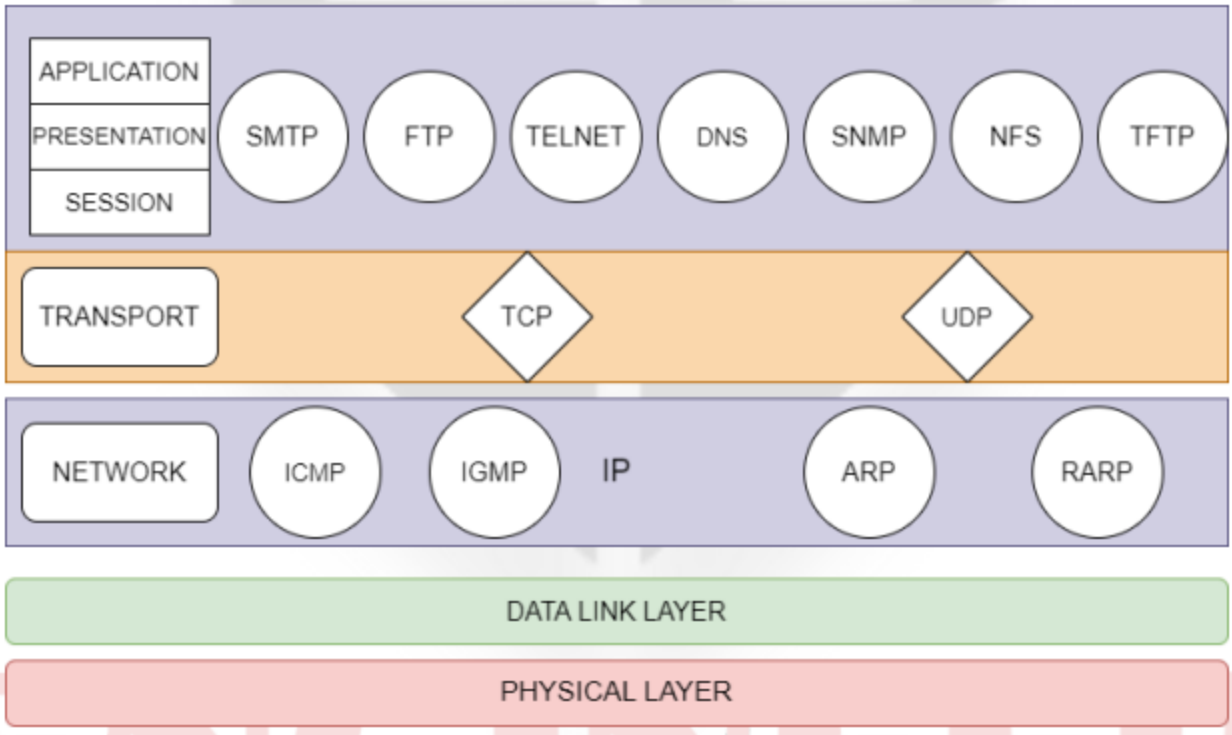
4. The Directory Services

OSI model acts as a reference model and is not implemented in the Internet because of its late invention. Current model being used is the TCP/IP model

TCP/IP MODEL

TCP/IP is a practical model that is used in the real world to transmit data over the internet and other networks. **TCP/IP protocol suite is made of five layers:** physical, data link, network, transport, and application OR

It is a four-layer model that includes the network interface layer, internet layer, transport layer, and application layer.



Network Interface Layer: The Network Interface Layer, also known as the Link Layer, is responsible for the physical transmission of data on a network. Some commonly used protocols at this layer include Ethernet, Wi-Fi, and Token Ring.

Network Layer TCP/IP supports the Internet working Protocol.

Four supporting protocols: **ARP, RARP, ICMP, and IGMP**

• **Internet Protocol (IP):** It is an unreliable and connectionless protocol(a best effort delivery service).The most widely used protocol at the Network Layer, IP provides a

mechanism for addressing and routing packets across the Internet and other IP-based networks. There are two versions of IP: IPv4 and IPv6.

• **Address Resolution Protocol:** is used to associate a logical address with a physical address.

• **Reverse Address Resolution Protocol:** allows a host to discover its Internet address when it knows only its physical address

• **Internet Control Message Protocol:** sends query and error reporting messages.

• **Internet Group Message Protocol:** is used to facilitate the simultaneous transmission of a message to a group of recipients.

Logical addresses

A logical address in the Internet is currently a 32-bit address that can uniquely define a host connected to the Internet. No two publicly addressed and visible hosts on the Internet can have the same IP address.

Transport Layer

Transport layer is represented by two protocols:

TCP and UDP

TCP is a connection-oriented protocol

o Does not mean it has a physical connection between sender and receiver

o TCP provides the function to allow a connection virtually exists – also called virtual circuit

UDP provides the functions:

o Dividing a chunk of data into segments

o Reassembly segments into the original chunk

o Provide further the functions such as reordering and data resend. Port addresses The label assigned to a process is called a port address.

A port address in TCP/IP is 16 bits in length.

Example:

• FTP: 20

• Telnet: 23

• HTTP: 80

• DNS: 53

• HTTPs: 443

Application Layer

The application layer in TCP/IP is equivalent to the combined session, presentation, and application layers in the OSI model.

Application layer protocols define the rules when implementing specific network applications Rely on the underlying layers to provide accurate and efficient data delivery

Typical protocols:

o Telnet – Remote terminal protocol For remote login on any other computer on the network

o SMTP – Simple Mail Transfer Protocol For mail transfer

o HTTP – Hypertext Transfer Protocol For Web browsing

o TFTP - Trivial File Transfer Protocol

o FTP – File Transfer Protocol For file transfer

TFTP is a simple, lightweight protocol that is primarily used for transferring small files over a network, while FTP is a more robust protocol that can be used for transferring larger files and includes more advanced security and error handling features.

What are design issues of layers?

There are some key design issues occur in computer networks are present in several layers

Addressing: Every layer needs a mechanism to identify senders and receivers.

Error control: It's an important issue because physical communication circuits are not perfect. Many error detecting and error correcting codes are available. Both sending and receiving ends must agree to use any one code.

Flow control: This property leads to mechanisms for disassembling, transmitting and then reassembling messages.

Multiplexing & Demultiplexing: These two are must for improving the n/w system.

Routing: When there are multiple paths between source and destination, a route must be chosen

Comparison of OSI and TCP/IP Network Model

OSI is a generic, protocol independent standard, acting as a communication gateway between the network and end user	The TCP/IP model is based on standard protocols around which the Internet has developed.
In the OSI model the transport layer guarantees the delivery of packets.	The TCP/IP model is based on standard protocols around which the Internet has developed.
Following the horizontal approach OSI model has a separate Presentation layer and Session layer.	The TCP/IP model is based on standard protocols around which the Internet has developed.

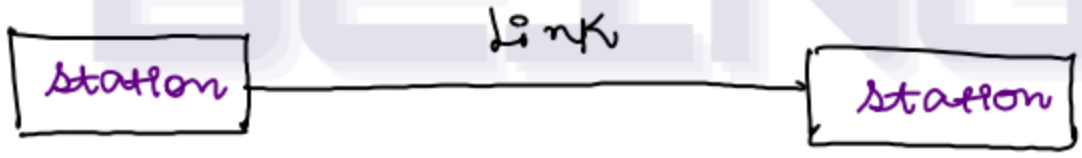
Transport Layer is Connection Oriented Network Layer is both Connection Oriented and Connection less OSI is a reference model around which the networks are built.	In the TCP/IP model the transport layer does not guarantees delivery of packets. Still the TCP/IP model is more reliable.
Generally it is used as a guidance tool.	Follows a vertical approach. TCP/IP does not have a separate Presentation layer or Session layer.
	Transport Layer is both Connection Oriented and Connection less. Network Layer is Connection less. TCP/IP model is, in a way implementation of the OSI model

Network Topologies

Type of Connection A network is two or more devices connected through links. A link is a communications pathway that transfers data from one device to another.

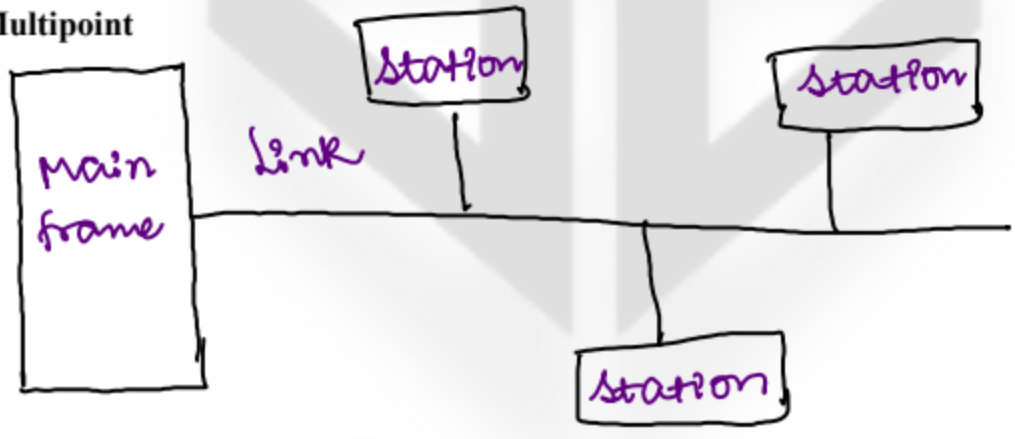
There are two possible types of connections: point-to-point and multipoint.

Point-to-Point



A point-to-point connection provides a dedicated link between two devices. The entire capacity of the link is reserved for transmission between those two devices. Most point-to-point connections use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links, are also possible When you change television channels by infrared remote control, you are establishing a point-to-point connection between the remote control and the television's control system.

Multipoint



A multipoint (also called multi-drop) connection is one in which more than two specific devices share a single link In a multipoint environment, the capacity of the channel is shared, either spatially or temporally. If several devices can use the link simultaneously, it is a spatially shared connection. If users must take turns, it is a timeshared connection

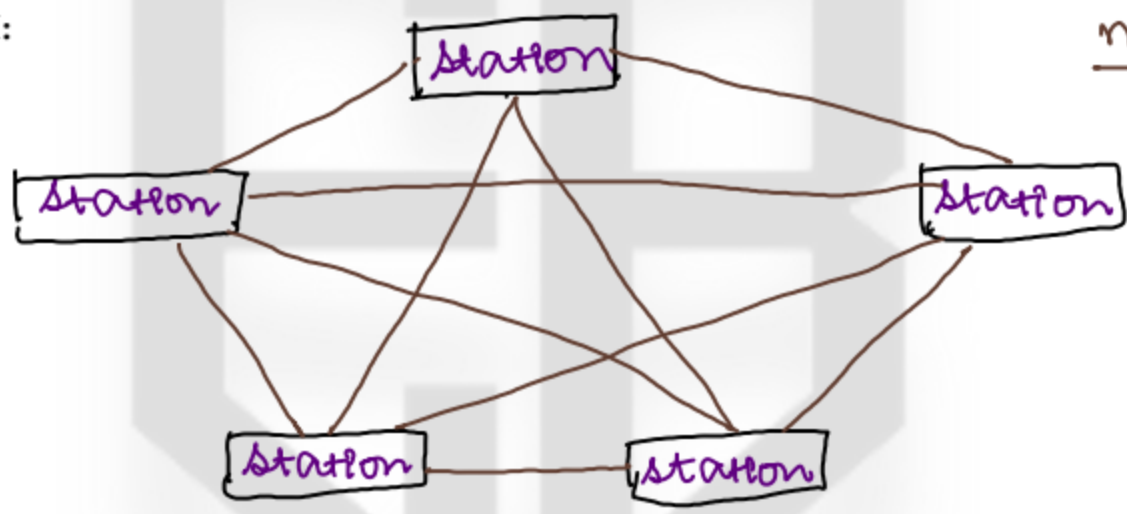
Network topology

In computer networking, network topology refers to the physical or logical layout of the devices, nodes, and links that make up a network. It describes the way in which nodes and devices are interconnected with each other and how data is transmitted between them.

There are several types of network topologies, including:

- **Bus Topology**
- **Star Topology**
- **Ring Topology**
- **Mesh Topology**

MESH:



A mesh topology is the one where every node is connected to every other node in the network. . The number of connections in this network can be calculated using the following formula (n is the number of computers in the network): n(n-1)/2

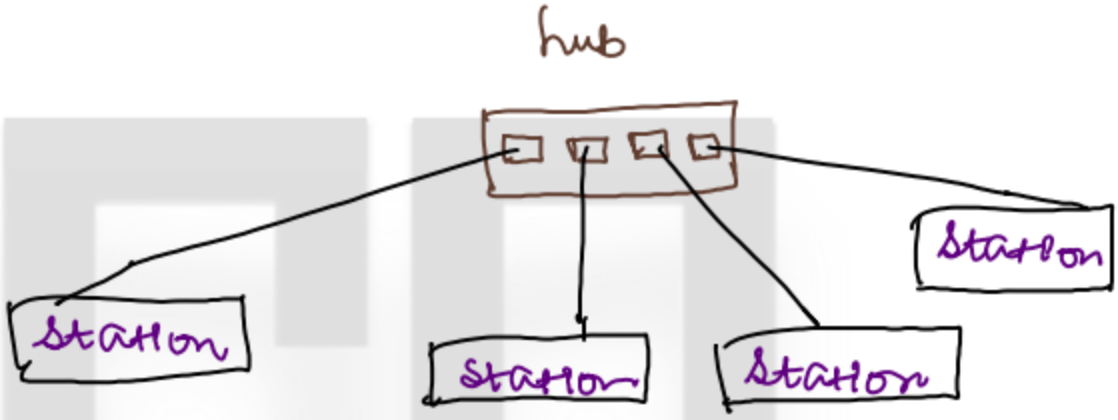
Advantages of a mesh topology

- Can handle high amounts of traffic, because multiple devices can transmit data simultaneously.
- A failure of one device does not cause a break in the network or transmission of data.
- Adding additional devices does not disrupt data transmission between other devices.

Disadvantages of a mesh topology

- The cost to implement is higher than other network topologies, making it a less desirable option.
- Building and maintaining the topology is difficult and time consuming.
- The chance of redundant connections is high, which adds to the high costs and potential for reduced efficiency.

Star Topology



A star network, star topology, is one of the most common network setups. In this configuration, every node connects to a central network device, like a hub, switch, or computer. The central network device acts as a server and the peripheral devices act as clients. a coaxial cable or a RJ-45 network cable is used to connect computers together.

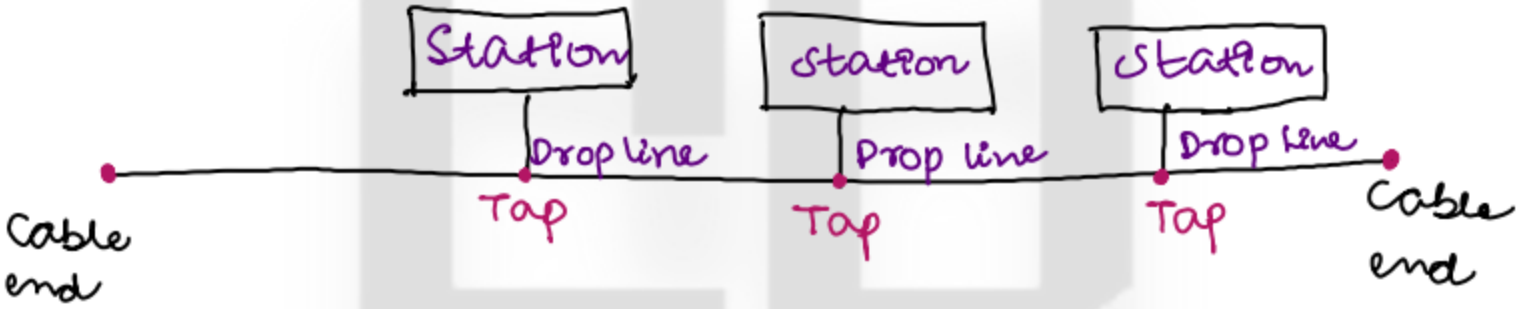
Advantages of star topology

- Centralized management of the network, through the use of the central computer, hub, or switch. Easy to add another computer to the network.
- If one computer on the network fails, the rest of the network continues to function normally.
- The star topology is used in local-area networks (LANs), High-speed LANs often use a star topology with a central hub.

Disadvantages of star topology

- Can have a higher cost to implement, especially when using a switch or router as the central network device.
- The central network device determines the performance and number of nodes the network can handle.
- If the central computer, hub, or switch fails, the entire network goes down and all computers are disconnected from the network

Bus Topology



A line topology, a bus topology is a network setup in which each computer and network device are connected to a single cable or backbone.

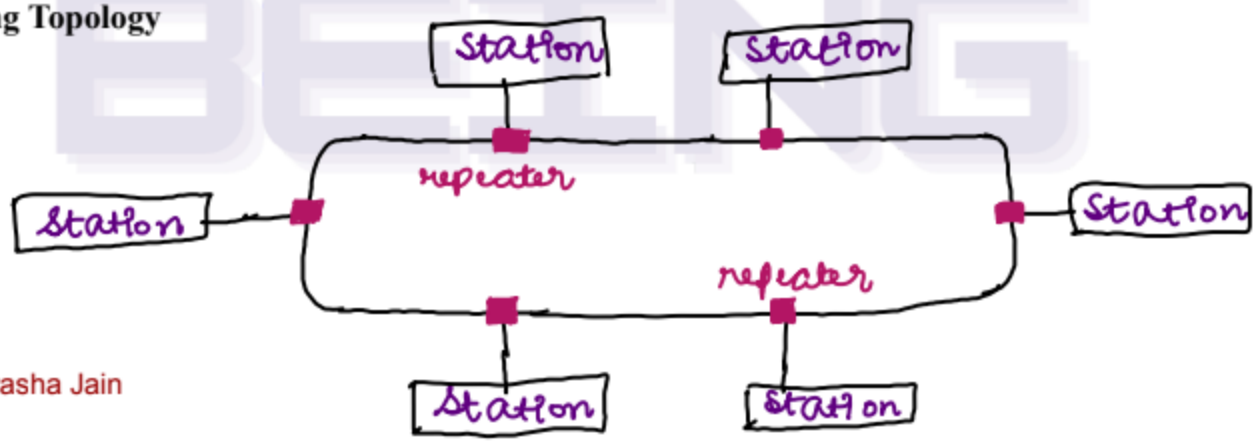
Advantages of bus topology

- It works well when you have a small network.
- It's the easiest network topology for connecting computers or peripherals in a linear fashion.
- It requires less cable length than a star topology.

Disadvantages of bus topology

- It can be difficult to identify the problems if the whole network goes down.
- It can be hard to troubleshoot individual device issues.
- Bus topology is not great for large networks.
- Terminators are required for both ends of the main cable.
- Additional devices slow the network down.
- If a main cable is damaged, the network fails or splits into two.

Ring Topology



A ring topology is a network configuration in which device connections create a circular data path. In a ring network, packets of data travel from one device to the next until they reach their destination. Most ring topologies allow packets to travel only in one direction, called a unidirectional ring network. Others permit data to move in either direction, called bidirectional.

Advantages of ring topology

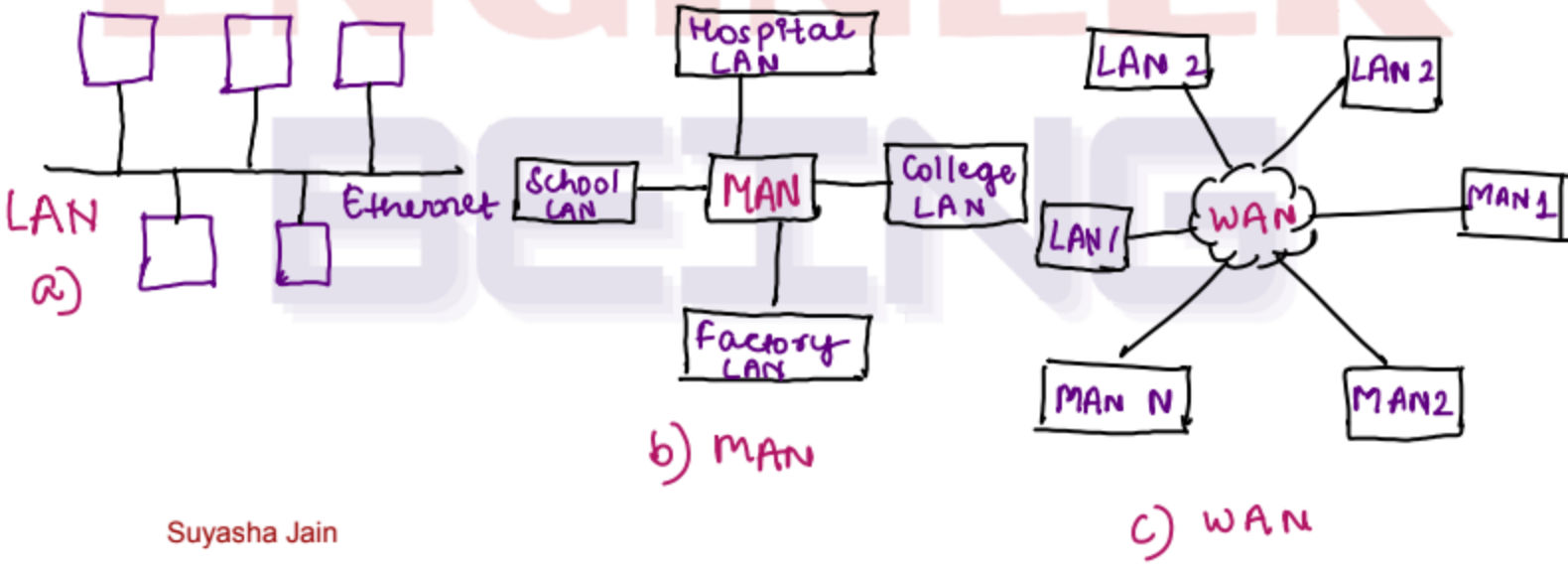
- All data flows in one direction, reducing the chance of packet collisions.
- A network server is not needed to control network connectivity between each workstation.
- Data can transfer between workstations at high speeds.
- Additional workstations can be added without impacting performance of the network

Disadvantages of ring topology

- All data being transferred over the network must pass through each workstation on the network, which can make it slower than a star topology.
- The entire network will be impacted if one workstation shuts down.
- The hardware needed to connect each workstation to the network is more expensive than Ethernet cards and hubs/switches.

Types of Network based on size

The three primary network categories are LAN, WAN and MAN. Each network differs in their characteristics such as distance, transmission speed, cables and cost.



LAN (Local Area Network) Group of interconnected computers within a small area. (room, building, campus) Two or more pc's can form a LAN to share files, folders, printers, applications and other devices. Coaxial or CAT 5 cables are normally used for connections. Due to short distances, errors and noise are minimal. Data transfer rate is 10 to 100 mbps. Example: A computer lab in a school.

MAN (Metropolitan Area Network) Design to extend over a large area. Connecting a number of LAN's to form a larger network, so that resources can be shared. Networks can be up to 5 to 50 km. Owned by organization or individual. Data transfer rate is low compared to LAN. Example: Organization with different branches located in the cit

WAN (Wide Area Network) Are country and worldwide networks. Contains multiple LAN's and MAN's. Distinguished in terms of geographical range. Uses satellites and microwave relays. Data transfer rate depends upon the ISP provider and varies over the location. Best example is the internet.

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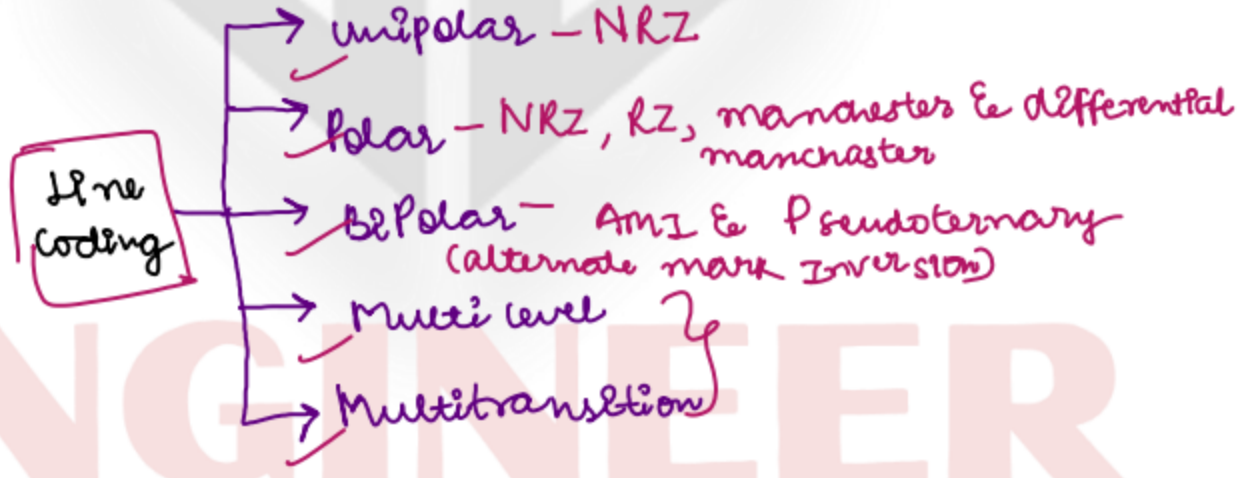
Digital Transmission

Ques : Discuss encoding types in the physical layer of ISO-OSI model. **2021-22 10M**

Ques : Encode the data-stream 10011010 using the following encoding scheme:
(i) Unipolar (ii) Bipolar NRZ-L (iii) Bipolar NRZ-I (iv) RZ (v) Manchestor (vi) Differential Manchester (vii) AMI
2018-19 10M

DIGITAL-TO-DIGITAL CONVERSION

How we can represent digital data by using digital signals.
The conversion involves three techniques: line coding, block coding, and scrambling. Line coding is always needed; block coding and scrambling may or may not be needed.



Unipolar

→ Show signal on one side of axis



Amplitude 10101



Polar - NRZ-L(non return to zero level) & NRZ-I (non return to zero inversion)

Signal on both sides of axis
- 1 with negative level
- 0 with positive level

1001

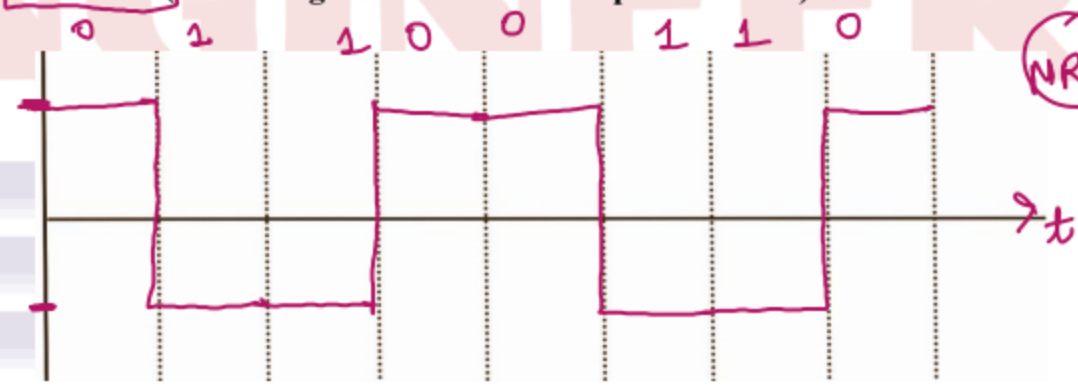


NRZ L

Bit duration

Polar NRZ-I 01100110 (1 with negative level & 0 with positive level)

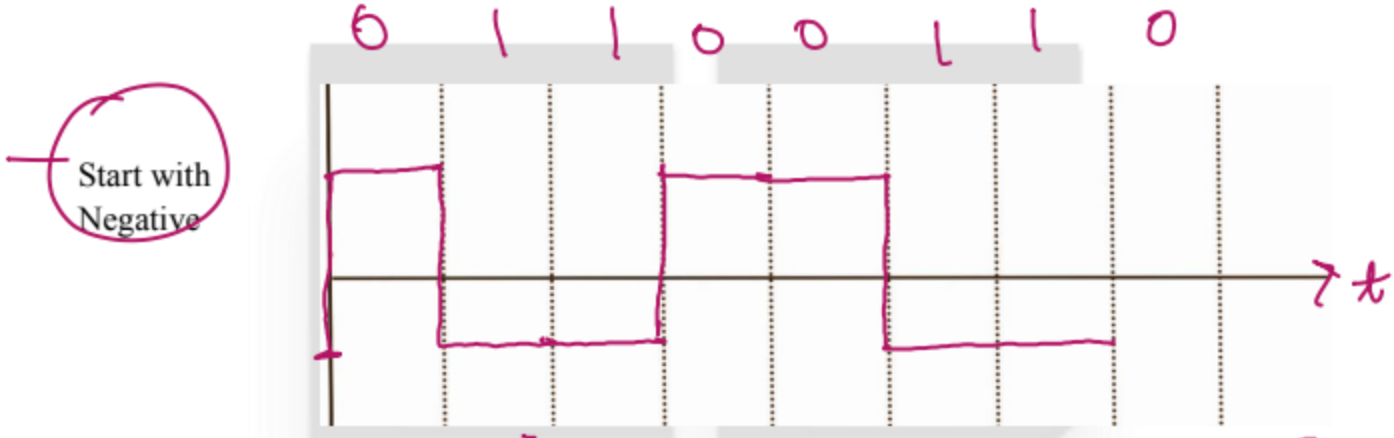
1 → -ve
0 → +ve



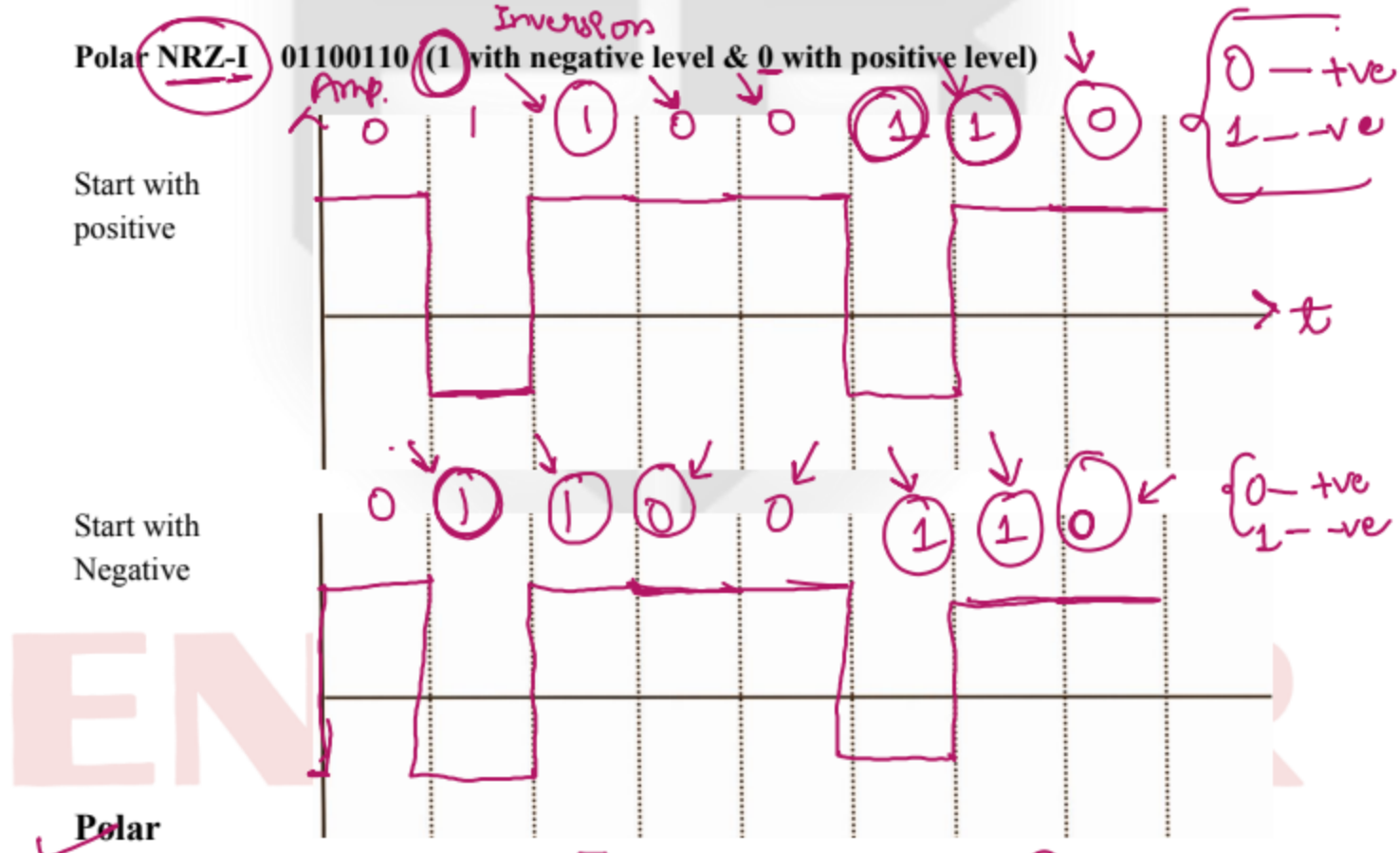
Start with positive

ENGINEER

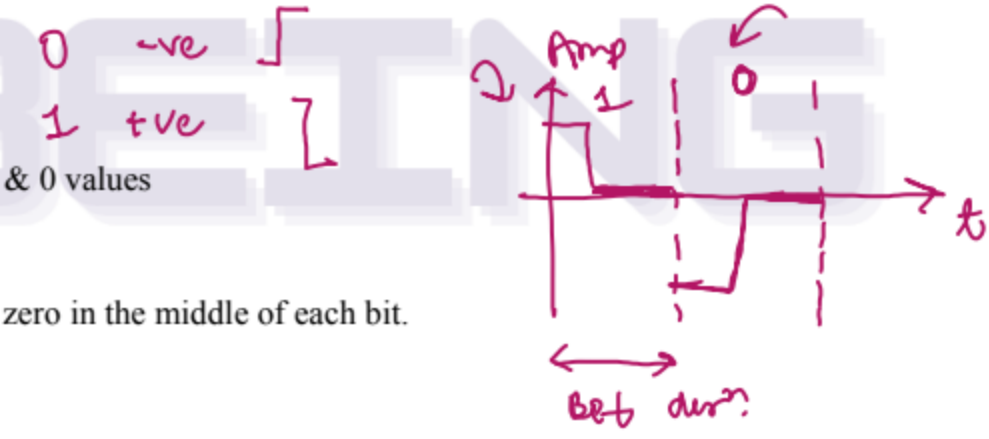
1 — -ve
0 — +ve



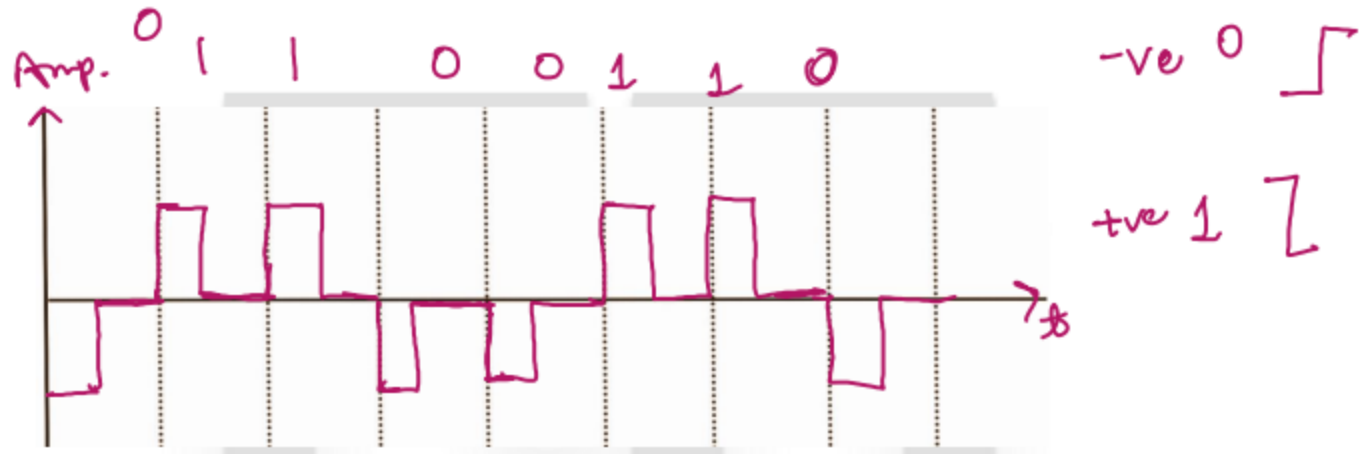
Polar NRZ-I 01100110 (1 with negative level & 0 with positive level)



Polar RZ(return to zero) Scheme



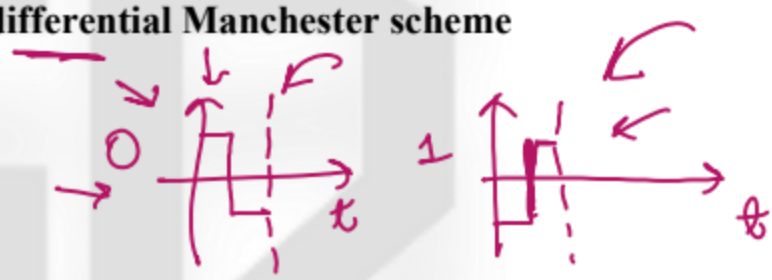
Positive , negative & 0 values
0 at negative side
1 at positive side
The signal goes to zero in the middle of each bit.



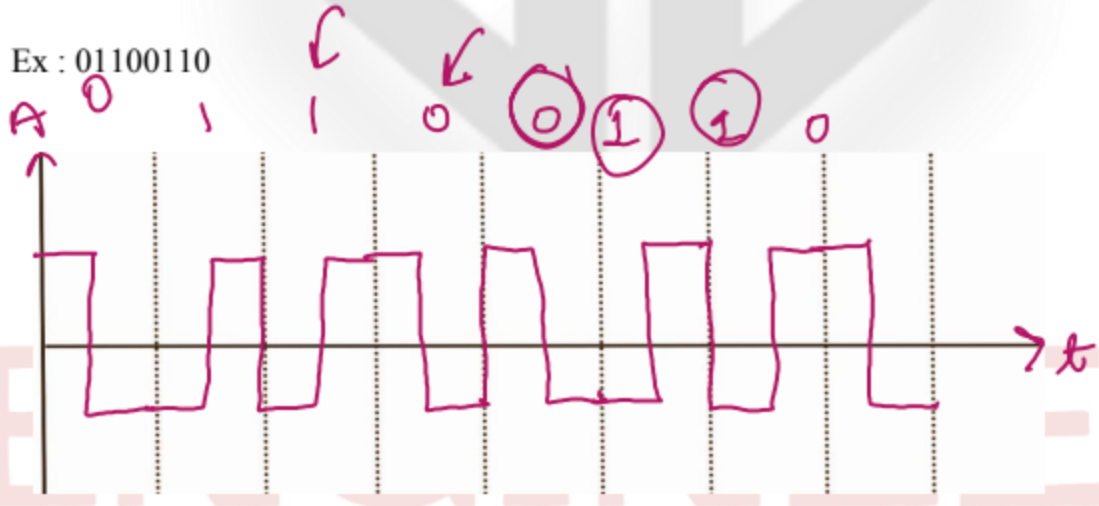
Polar biphase: Manchester and differential Manchester scheme

Duration bit is divided in two halves.
0 is represented as

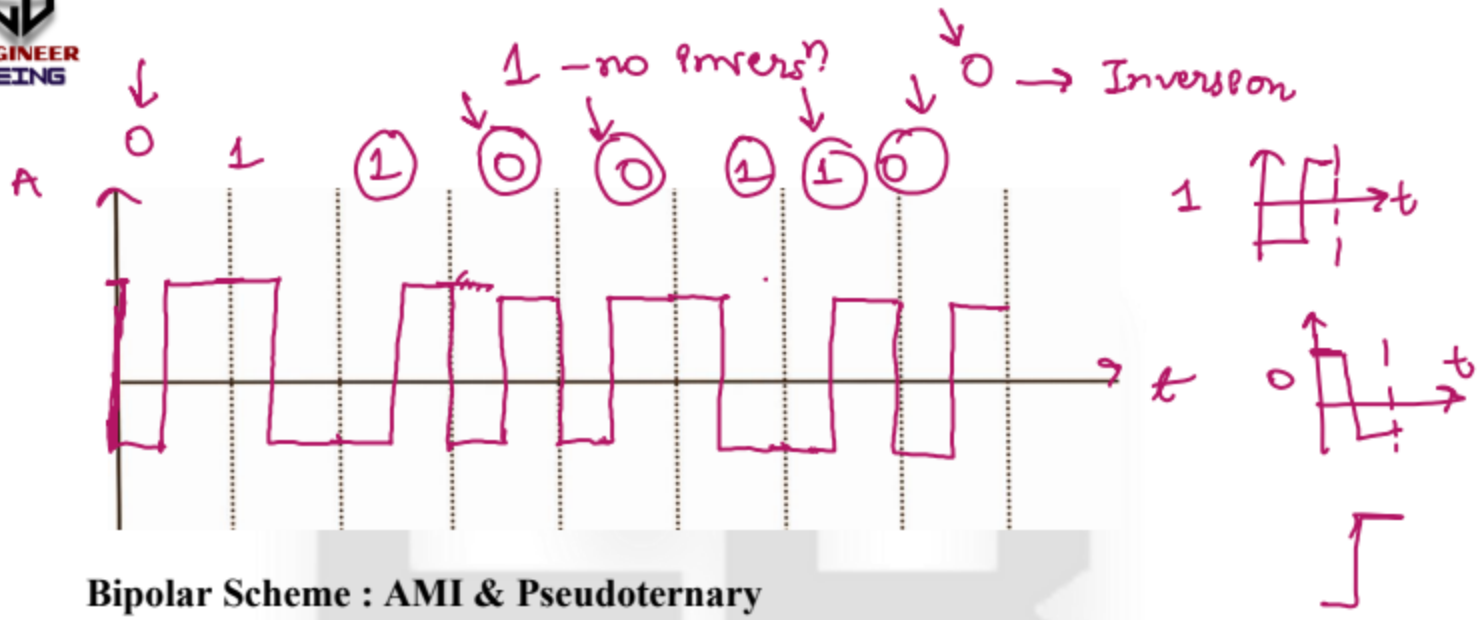
1 is represented as



Ex : 01100110



Differential Manchester 01100110
1 - no inversion
0 - inversion



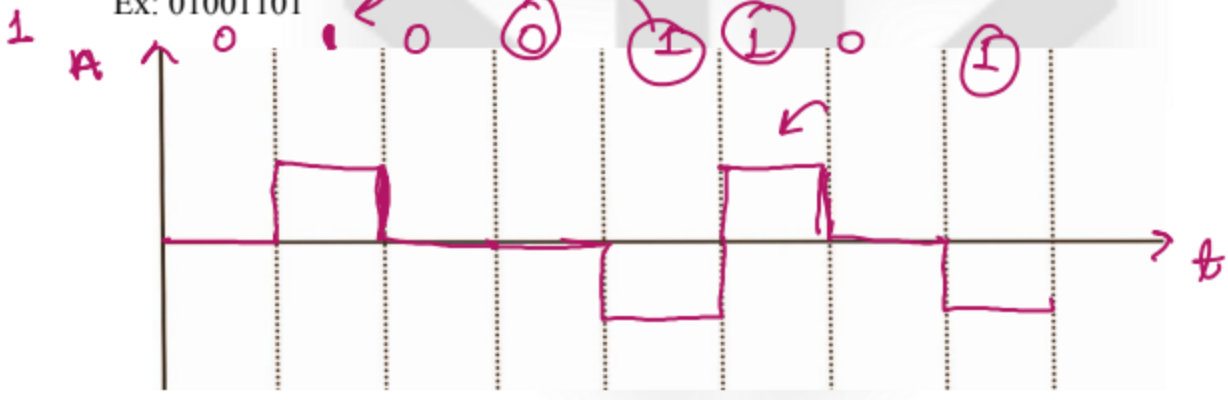
Bipolar Scheme : AMI & Pseudoternary

Three values positive , negative & zero

AMI

1 is represented alternatively in positive and negative direction & zero on x axis

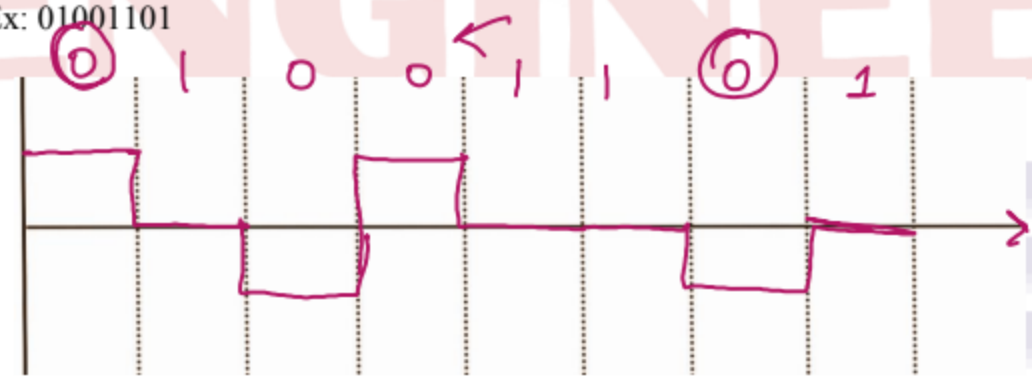
Ex: 01001101



Pseudoternary

0 is represented alternatively in positive and negative direction & 1 on x axis

Ex: 01001101



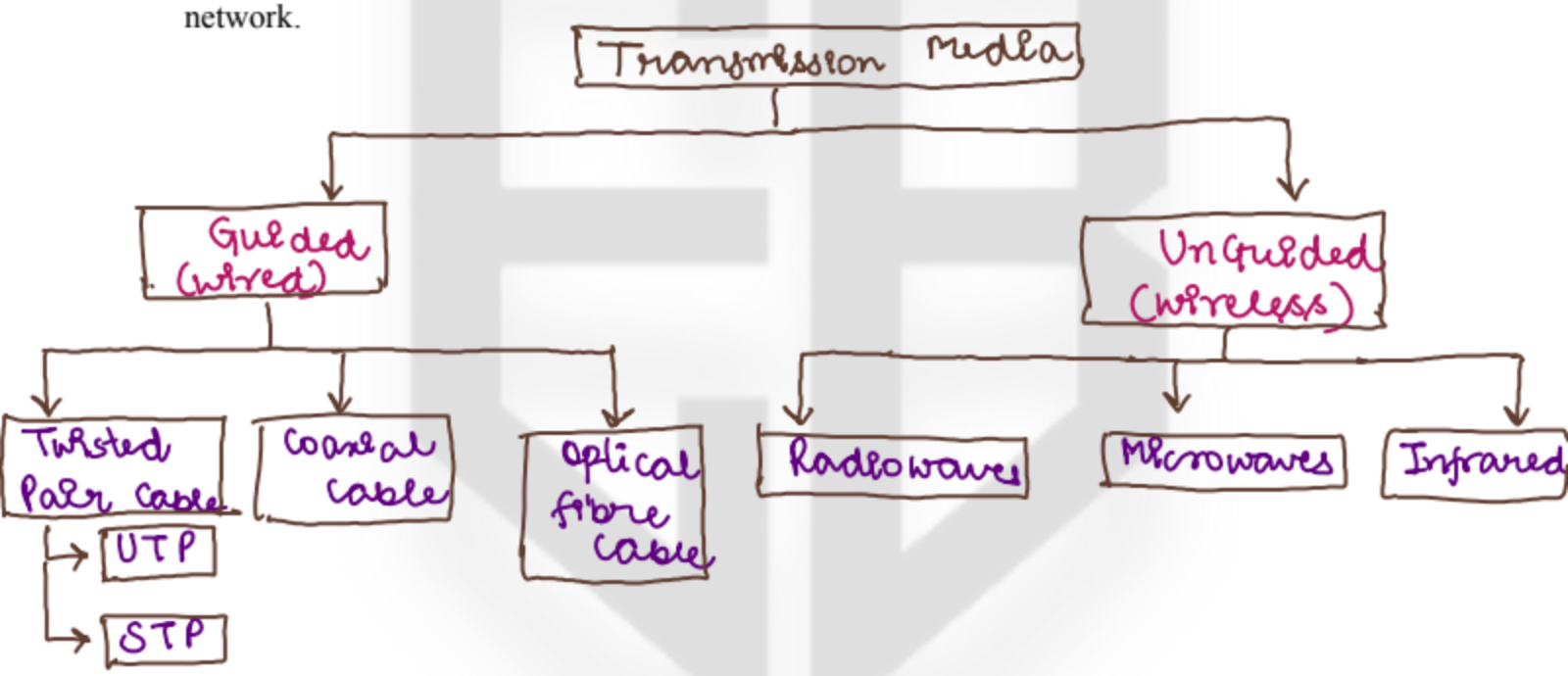
Composite Signals:

ENGINEER

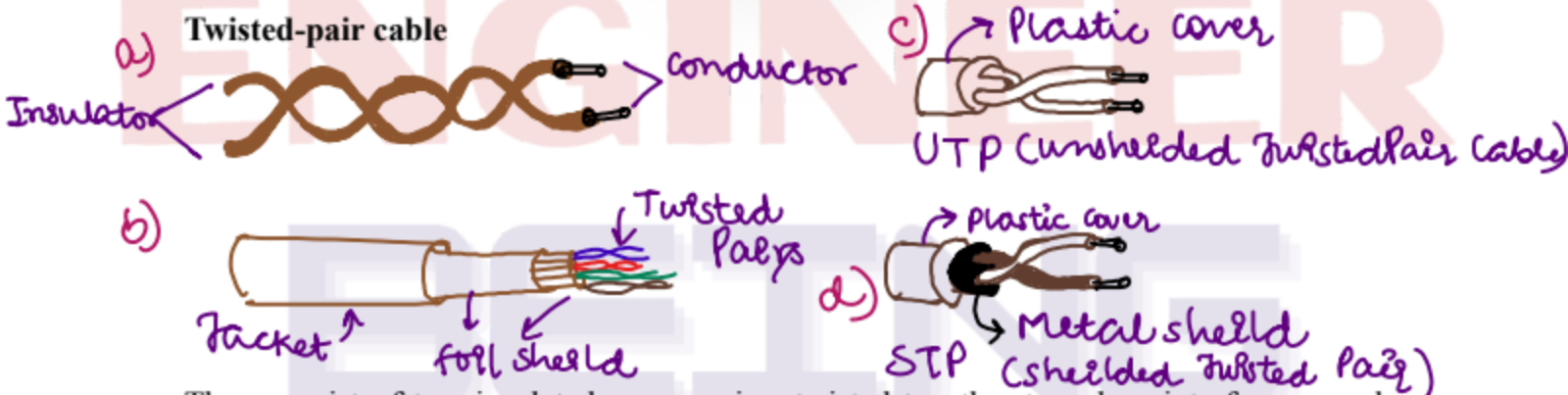


Transmission Media

Transmission media, also known as communication channels, refers to the physical pathways through which data is transmitted from one device to another in a computer network.



Guided media refers to the type of transmission media in computer networking that uses physical wires or cables to transmit data signals from one device to another.

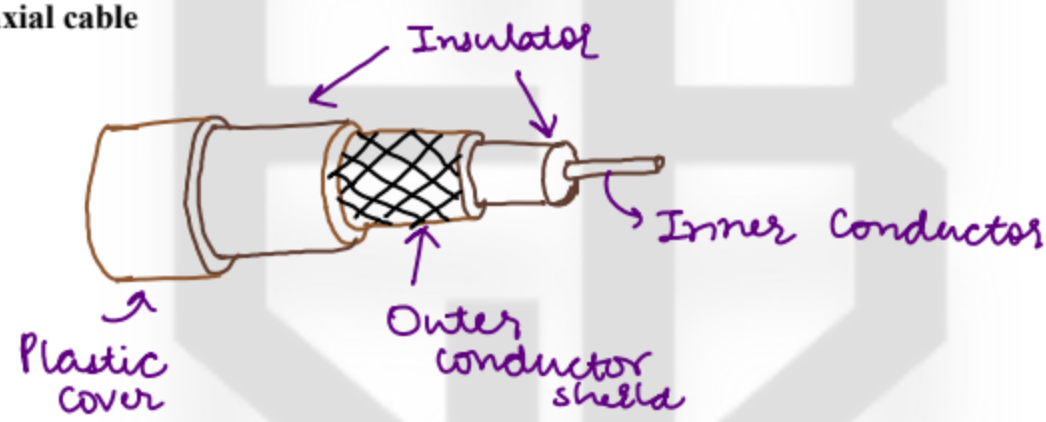


They consist of two insulated copper wires twisted together to reduce interference and crosstalk. Twisted-pair cables are used for Ethernet and other local area network (LAN) technologies.

Unshielded Versus Shielded Twisted-Pair Cable

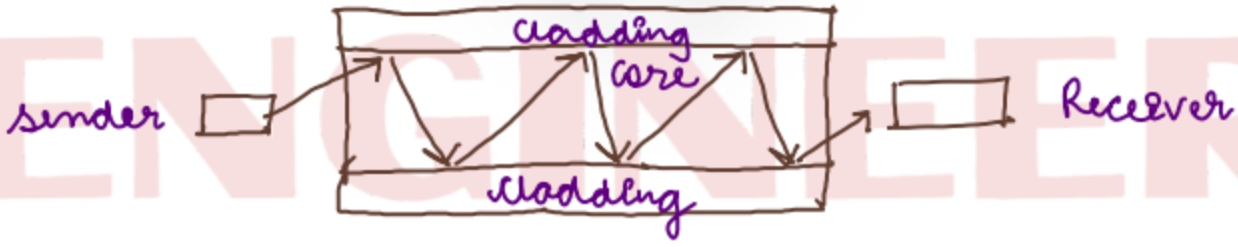
The most common twisted-pair cable used in communications is referred to as unshielded twisted-pair (UTP). STP cable has a metal foil or braided mesh covering that encases each pair of insulated conductors. Although metal casing improves the quality of cable by preventing the penetration of noise or crosstalk, it is bulkier and more expensive.

Coaxial cable



Coaxial cables consist of a single copper conductor surrounded by an insulating layer, a metallic shield, and an outer jacket. They are used for cable television (CATV) and high-speed data transmission. Cable TV networks also use coaxial cables.

Optical Fibre Cable



Fiber-Optic Cable A fiber-optic cable is made of glass or plastic and transmits signals in the form of light. Light travels in a straight line as long as it is moving through a single uniform substance. If a ray of light traveling through one substance suddenly enters another substance(of a different density), the ray changes direction. Optical fibers use reflection to guide light through a channel. A glass or plastic core is surrounded by a cladding of less dense glass or plastic.

Twisted pair cable	Co-axial cable	Optical fiber
1. Transmission of signals takes place in the electrical form over the metallic conducting wires.	1. Transmission of signals takes place in the electrical form over the inner conductor of the cable.	1. Signal transmission takes place in an optical forms over a glass fiber.
2. In this medium the noise immunity is low.	2. Coaxial having higher noise immunity than twisted pair cable.	2. Optical fiber has highest noise immunity as the light rays are unaffected by the electrical noise.
3. Twisted pair cable can be affected due to external magnetic field.	3. Coaxial cable is less affected due to external magnetic field.	3. Not affected by the external magnetic field.
4. Cheapest medium.	4. Moderate Expensive.	4. Expensive
5. Low Bandwidth.	5. Moderately high bandwidth.	5. Very high bandwidth
6. Attenuation is very high.	6. Attenuation is low.	6. Attenuation is very low.
7. Installation is easy.	7. Installation is fairly easy.	7. Installation is difficult.

UNGUIDED MEDIA: WIRELESS

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication.

Radio Waves



Suyasha Jain

- Radio Waves Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called radio waves.
- Radio waves are omni directional. When an antenna transmits radio waves, they are propagated in all directions. This means that the sending and receiving antennas do not have to be aligned. A sending antenna sends waves that can be received by any receiving antenna.
- The Omni directional characteristics of radio waves make them useful for multicasting, in which there is one sender but many receivers.
- AM and FM radio, television, maritime radio, cordless phones, and paging are examples of multicasting.

Microwaves



- Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves. Microwaves are unidirectional. The sending and receiving antennas need to be aligned. The unidirectional property has an obvious advantage.
- A pair of antennas can be aligned without interfering with another pair of aligned antennas
- Unidirectional Antenna Microwaves need unidirectional antennas that send out signals in one direction. Two types of antennas are used for microwave communications: the parabolic dish and the horn Applications:
- Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LAN

Infrared

- Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication. Infrared waves, having high frequencies, cannot penetrate walls.
- This advantageous characteristic prevents interference between one system and another; a short range communication system in one room cannot be affected by another system in the next room. When we use our infrared remote control, we do not interfere with the use of the remote by our neighbors.
- Infrared signals are useless for long-range communication. In addition, we cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication.

Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.

Causes of Transmission Impairment:

o Attenuation: Attenuation means the loss of energy, i.e., the strength of the signal decreases with increasing the distance which causes the loss of energy.

o Distortion: Distortion occurs when there is a change in the shape of the signal. This type of distortion is examined from different signals having different frequencies. Each frequency component has its own propagation speed, so they reach at a different time which leads to the delay distortion.

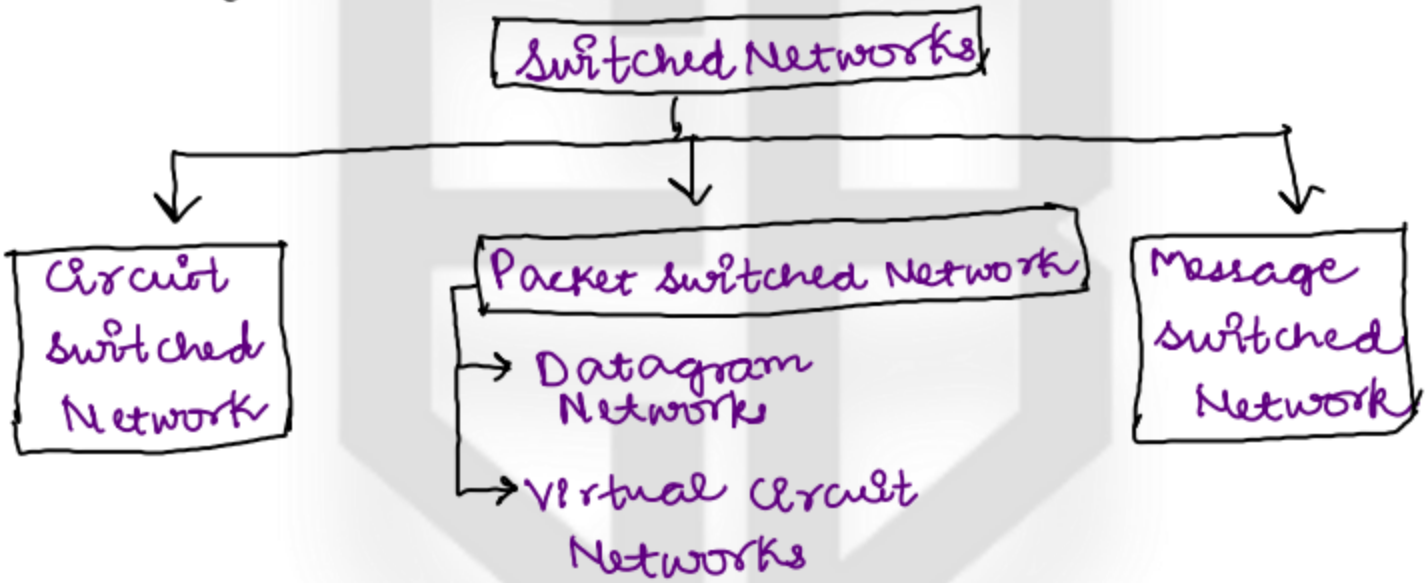
o Noise: When data is traveled over a transmission medium, some unwanted signal is added to it which creates the noise.

Switching

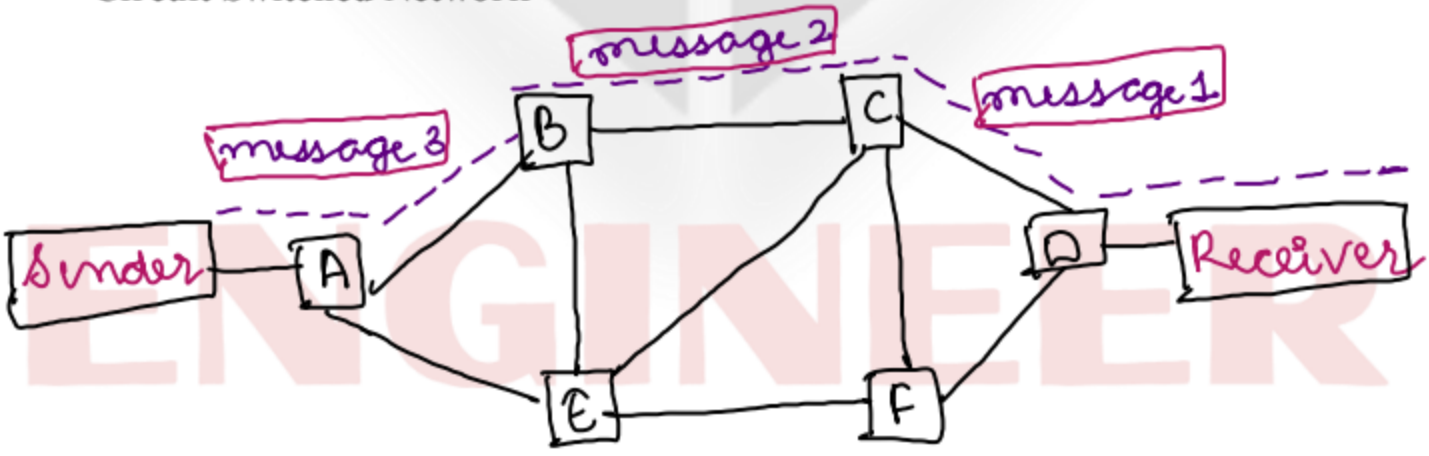
A switched network consists of a series of interlinked nodes, called switches. Switches are devices capable of creating temporary connections between two or more devices linked to the switch. In a switched network, some of these nodes are connected to the end systems (computers or telephones, for example). Others are used only for routing.

Switching Techniques

In large networks, there may be more than one path for transmitting data from sender to receiver. Selecting a path that data must take out of the available options is called switching.



Circuit Switched Network



A circuit-switched network consists of a set of switches connected by physical links. A connection between two stations is a dedicated path made of one or more links

Three phases:

- Connection setup means creating dedicated channels between the switches.
- Data Transfer Phase : After the establishment of the dedicated circuit (channels), the two parties can transfer data.
- Teardown Phase: When one of the parties needs to disconnect, a signal is sent to each switch to release the resources.

Advantages:

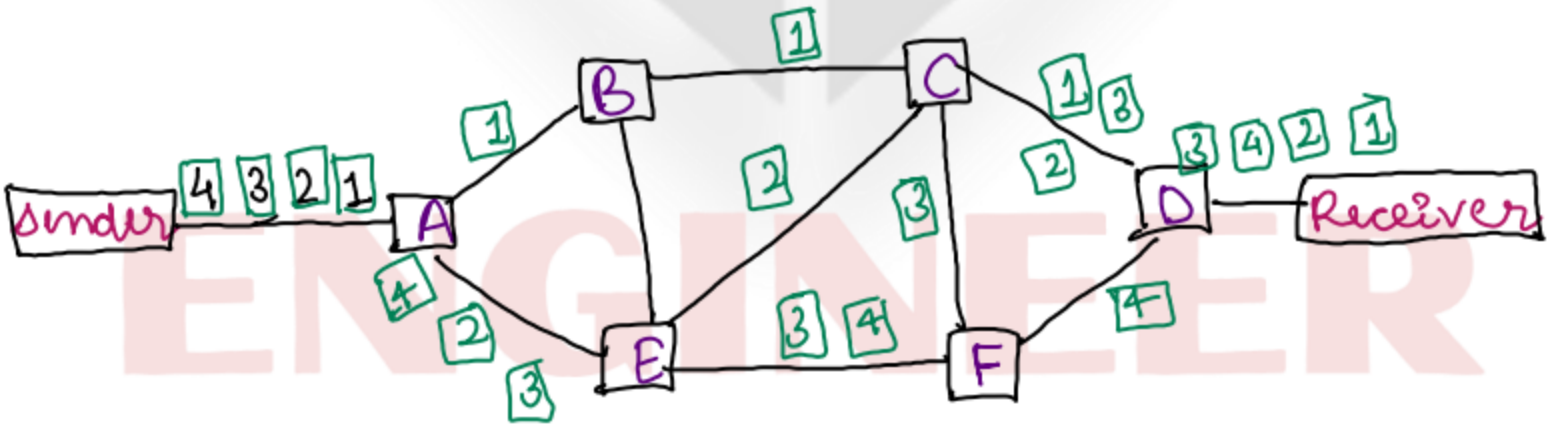
❑ The communication channel (once established) is dedicated.

Disadvantages:

❑ Possible long wait to establish a connection, (10 seconds, more on long-distance or international calls.) during which no data can be transmitted.

❑ More expensive than any other switching techniques, because a dedicated path is required for each connection.

Packet switching



Packet switching is a connectionless network switching technique. Here, the message is divided and grouped into a number of units called packets that are individually routed from the source to the destination. There is no need to establish a dedicated circuit for communication.

Packet Switching: Datagram

❑ Datagram packet switching is similar to message switching in that each packet is a self contained unit with complete addressing information attached.

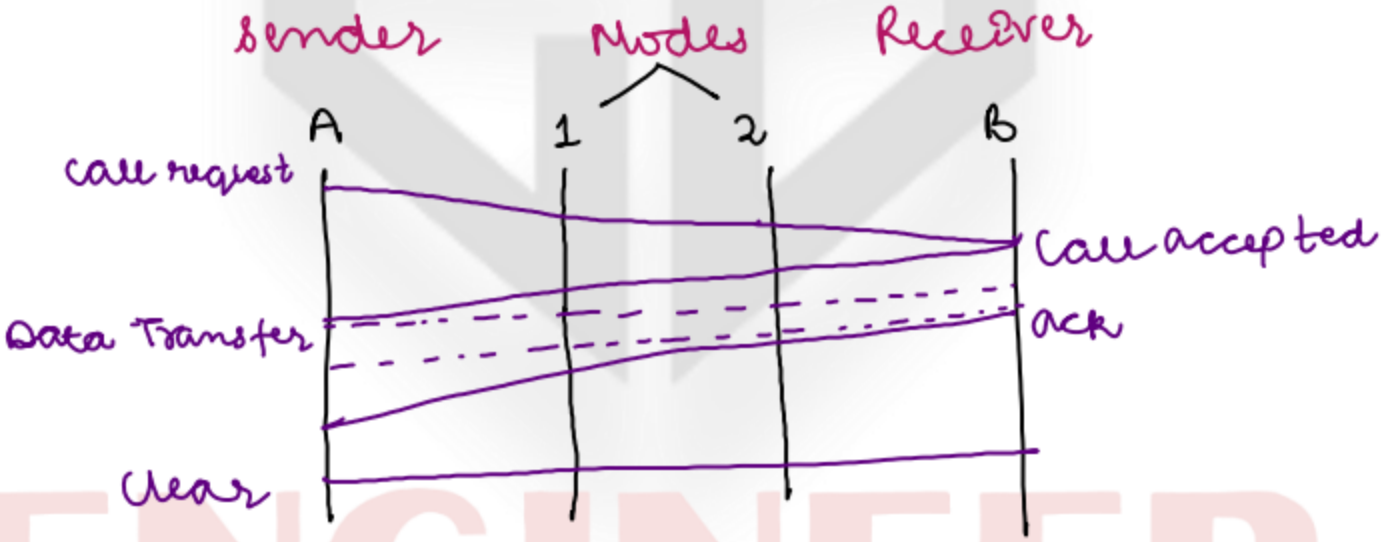
❑ This fact allows packets to take a variety of possible paths through the network.

❑ So the packets, each with the same destination address, do not follow the same route, and they may arrive out of sequence at the exit point node (or the destination).

❑ Reordering is done at the destination point based on the sequence number of the packets.

❑ It is possible for a packet to be destroyed if one of the nodes on its way crashes momentarily. Thus all its queued packets may be lost.

Packet Switching: Virtual Circuit



A virtual-circuit network is normally implemented in the data link layer, while a circuit-switched network is implemented in the physical layer and a datagram network in the network layer

In the virtual circuit approach, a preplanned route is established before any data packets are sent.

❑ A logical connection is established when

➢ A sender send a "call request packet" to the receiver and

➢ The receiver send back an acknowledge packet "call accepted packet" to the sender if the receiver agrees on conversational parameters.

Virtual circuits imply acknowledgements, flow control, and error control, so virtual circuits are reliable

❑ Virtual Circuits offer guarantees that

➢ the packets sent arrive in the order sent

➢ with no duplicates

➢ with no errors regardless of how they are implemented internally

Issue	Datagram network	Virtual Circuit Network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated

Advantages of packet switching

❑ Packet switching is cost effective, because switching devices do not need a massive amount of secondary storage.

❑ Packet switching offers improved delay characteristics, because there are no long messages in the queue (maximum packet size is fixed).

❑ Packets can be rerouted if there is any problem, such as busy or disabled links.

❑ The advantage of packet switching is that many network users can share the same channel at the same time. Packet switching can maximize link efficiency by making optimal use of link bandwidth.

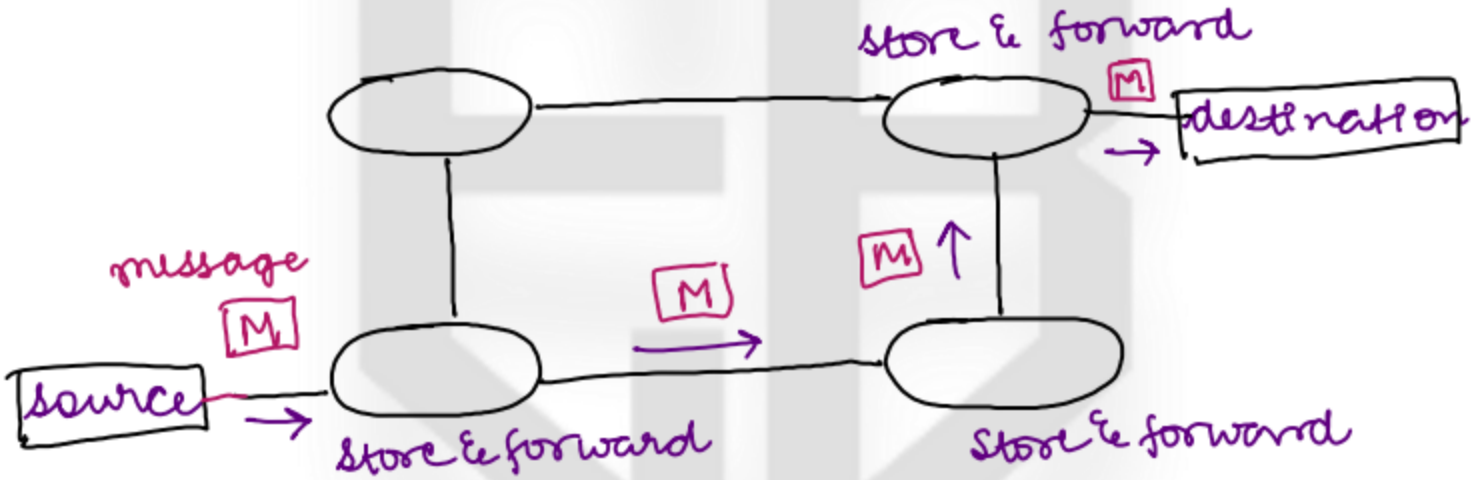
Disadvantages of packet switching

❑ Protocols for packet switching are typically more complex.

❑ It can add some initial costs in implementation.

❑ If packet is lost, sender needs to retransmit the data

Message Switching



❑ With message switching there is no need to establish a dedicated path between two stations.

❑ When a station sends a message, the destination address is appended to the message.

❑ The message is then transmitted through the network, in its entirety, from node to node.

❑ Each node receives the entire message, stores it in its entirety on disk, and then transmits the message to the next node.

❑ This type of network is called a store-and-forward network

Advantages:

❑ Channel efficiency can be greater compared to circuit-switched systems, because more devices are sharing the channel.

❑ Traffic congestion can be reduced, because messages may be temporarily stored in route.

❑ Message priorities can be established due to store-and-forward technique.

❑ Message broadcasting can be achieved with the use of a broadcast address appended in the message.

Disadvantages:

❑ Message switching is not compatible with interactive applications.

❑ Store-and-forward devices are expensive, because they must have large disks to hold potentially long messages.

Digital Transmission

Ques : Discuss encoding types in the physical layer of ISO-OSI model. **2021-22 10M**

Ques : Encode the data-stream 10011010 using the following encoding scheme:
(i) Unipolar (ii) Bipolar NRZ-L (iii) Bipolar NRZ-I (iv) RZ (v) Manchestor (vi) Differential Manchester (vii) AMI
2018-19 10M

DIGITAL-TO-DIGITAL CONVERSION

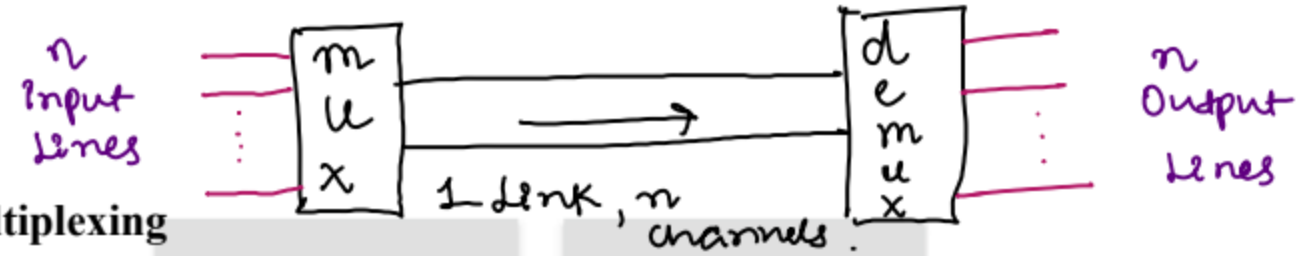
How we can represent digital data by using digital signals.
The conversion involves three techniques: line coding, block coding, and scrambling. Line coding is always needed; block coding and scrambling may or may not be needed

Unipolar

Show signal on one side of axis

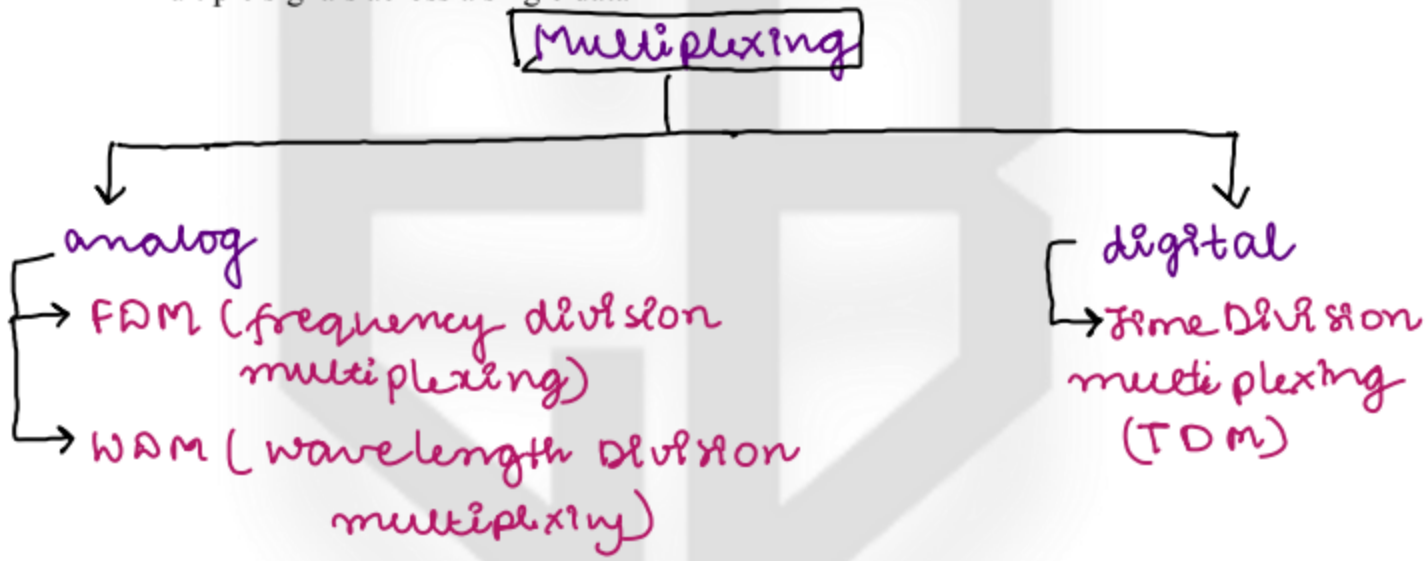
acts as an interface between the computer and the router or modem. NIC card is a layer 2 device which means that it works on both the physical and data link layers of the network model.

ENGINEER BEING



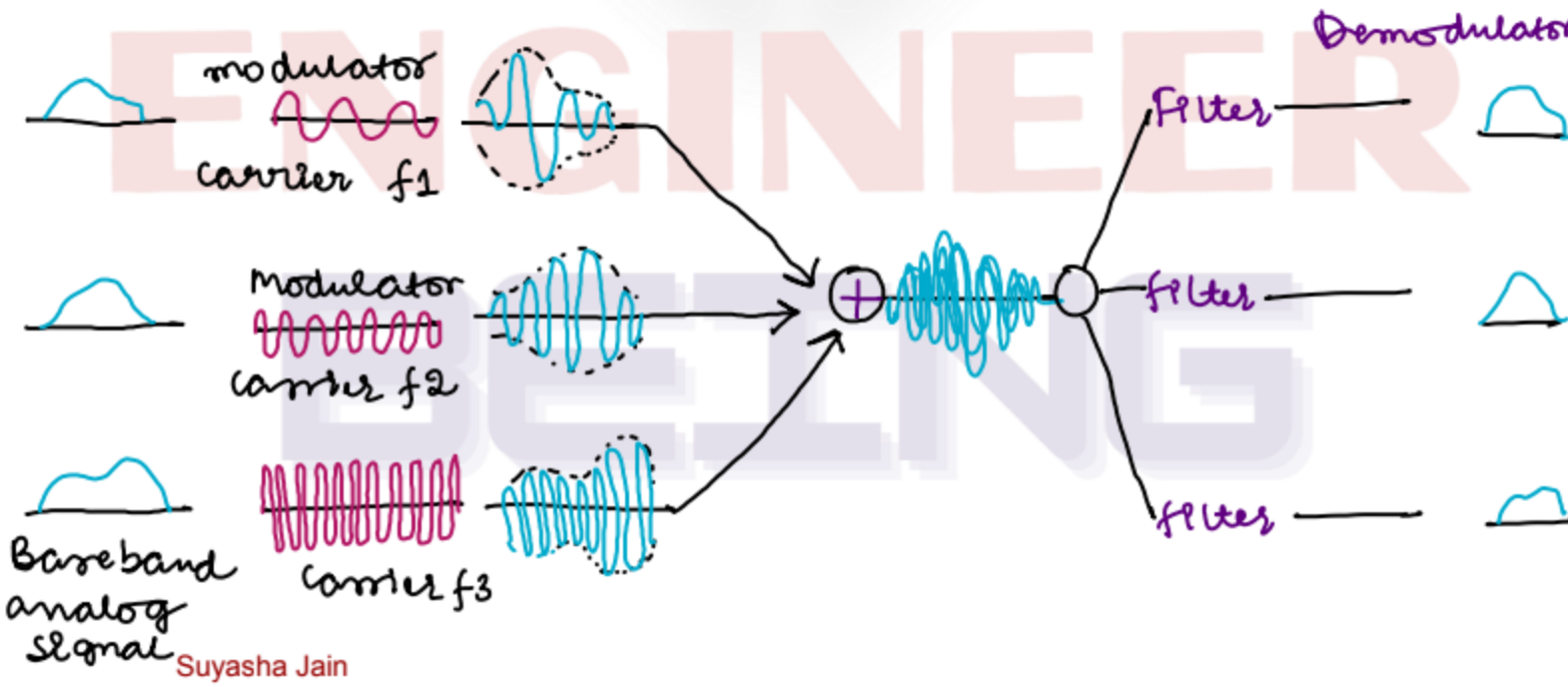
Multiplexing

- Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared.
- Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link



There are several types of multiplexing techniques, including:

Frequency-division multiplexing (FDM)

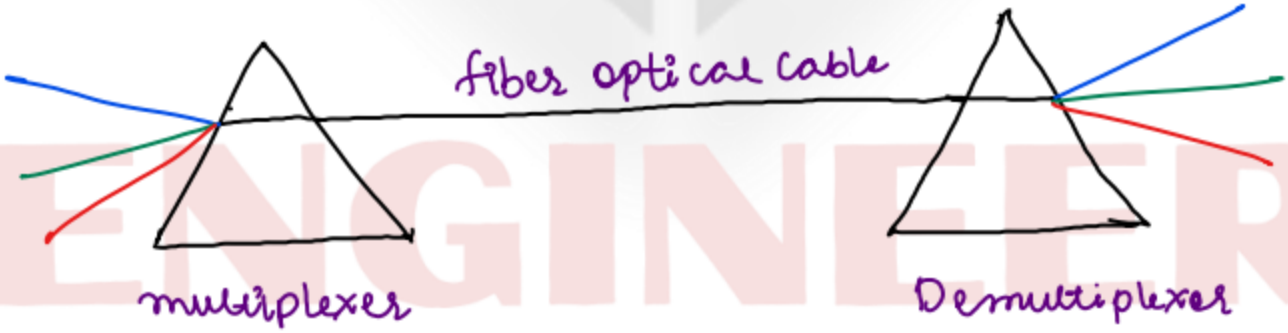


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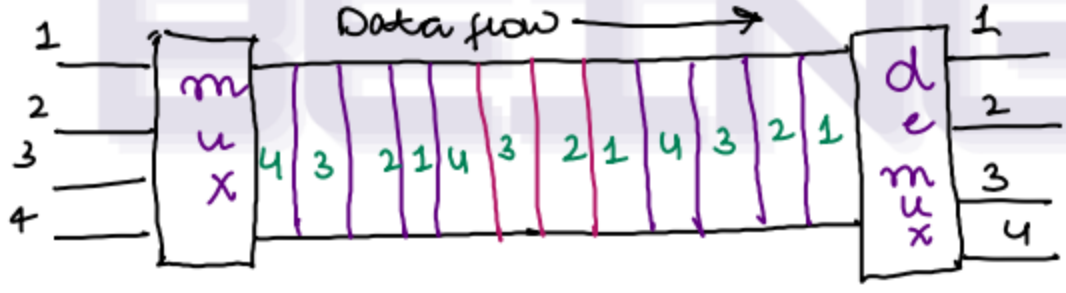
- The input signals are translated into frequency bands by using modulation techniques, and they are combined by a multiplexer to form a composite signal.
- The main aim of the FDM is to subdivide the available bandwidth into different frequency channels and allocate them to different devices.
- Using the modulation technique, the input signals are transmitted into frequency bands and then combined to form a composite signal.
- The carriers which are used for modulating the signals are known as sub-carriers. They are represented as f1,f2..fn.
- FDM is mainly used in radio broadcasts and TV networks.

Wavelength-division multiplexing (WDM)

- It is an analog multiplexing technique.
- Wavelength Division Multiplexing is the same as FDM except that the optical signals are transmitted through the fiber optic cable.
- WDM is used on fiber optics to increase the capacity of a single fiber.
- Prism can perform a role of multiplexer by combining the various optical signals to form a composite signal, and the composite signal is transmitted through a fiber optical cable.
- Prism also performs a reverse operation, i.e., demultiplexing the signal.



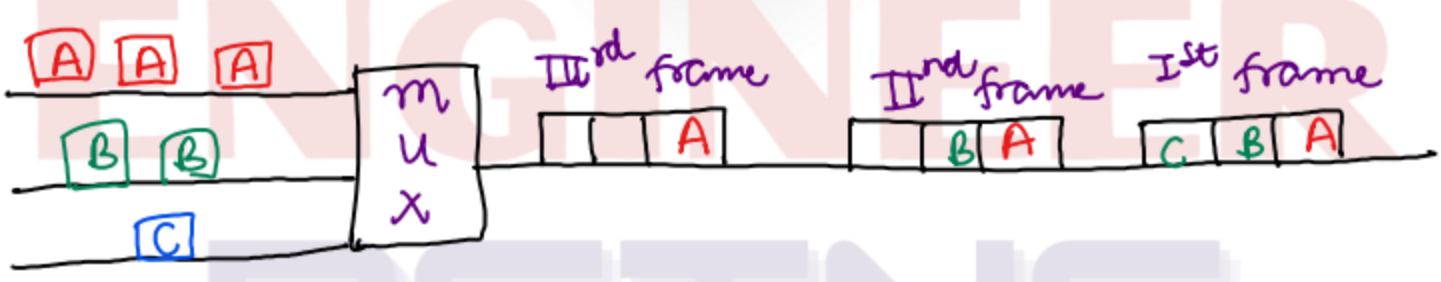
Time-division multiplexing (TDM):



- It is a digital technique.
- In Frequency Division Multiplexing Technique, all signals operate at the same time with different frequency, but in case of Time Division Multiplexing technique, all signals operate at the same frequency with different time.
- In Time Division Multiplexing technique, the total time available in the channel is distributed among different users. Therefore, each user is allocated with a different time interval known as a Time slot at which data is to be transmitted by the sender.
- A user takes control of the channel for a fixed amount of time.
- In Time Division Multiplexing technique, data is not transmitted simultaneously rather the data is transmitted one-by-one.

Synchronous Time Multiplexing

- A Synchronous TDM is a technique in which a time slot is preassigned to every device.
- In Synchronous TDM, each device is given some time slot irrespective of the fact that the device contains the data or not.
- If the device does not have any data, then the slot will remain empty.
- In Synchronous TDM, signals are sent in the form of frames. Time slots are organized in the form of frames. If a device does not have data for a particular time slot, then the empty slot will be transmitted.
- The most popular Synchronous TDM are T-1 multiplexing, ISDN multiplexing, and SONET multiplexing.



Asynchronous TDM

- An asynchronous TDM is also known as Statistical TDM.
- An asynchronous TDM is a technique in which time slots are not fixed as in the case of Synchronous TDM. Time slots are allocated to only those devices which have the data to send. Therefore, we can say that Asynchronous Time Division multiplexing transmits only the data from active workstations.
- An asynchronous TDM technique dynamically allocates the time slots to the devices.



Network Performance

Network performance is measured in two fundamental ways:

- Bandwidth (also called throughput) &
- Latency (also called delay)

The bandwidth of a network is given by the number of bits that can be transmitted over the network in a certain period of time.

Latency = propagation time + Transmission time + Queuing Time + Processing Delay

Propagation Time : Time required to travel from source to destination
= Distance/propagation speed11

Transmission time : The time required for transmission of message depends on the size Of message and bandwidth of the channel.
= Message Size/ Bandwidth

Dist / Prop speed

msg size / Bandwidth

Ques: What is the propagation time and transmission time for a 5 Mbyte message of bandwidth of the network is 1 Mbps. Distance is 12,000 kilometers and light travels at 2.4 into 10^8 m/sec.

Ques: If the bandwidth of the channel is 5 kbps, how long does it take 1,00,000 to pass?

ENGINEER BEING

Network Devices (Hub, Repeater, Bridge, Switch, Router, Gateways)

Repeaters: Repeaters are network devices that amplify and retransmit signals to extend their range. They are used to overcome signal degradation or attenuation that occurs when a signal travels over a long distance or through a medium that causes signal loss. Repeaters can be deployed in both wired and wireless networks and can be placed at various points in the network to extend its range. Overall, repeaters play a critical role in ensuring that data is transmitted reliably and efficiently in a computer network.

Hub: A hub is a basically multi-port repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. In other words, the collision domain of all hosts connected through Hub remains one. Also, they do not have the intelligence to find out the best path for data packets which leads to inefficiencies and wastage.

Active Hub:- These are the hubs that have their power supply and can clean, boost, and relay the signal along with the network. It serves both as a repeater as well as a wiring center. These are used to extend the maximum distance between nodes.

Passive Hub:- These are the hubs that collect wiring from nodes and power supply from the active hub. These hubs relay signals onto the network without cleaning and boosting them and can't be used to extend the distance between nodes.

Bridge – A bridge operates at the data link layer. A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of the source and destination. It is also used for interconnecting two LANs working on the same protocol.

Switch – A switch is a multiport bridge with a buffer and a design that can boost its efficiency(a large number of ports imply less traffic) and performance. A switch is a data link layer device. The switch can perform error checking before forwarding data, which makes it very efficient as it does not forward packets that have errors and forward good packets selectively to the correct port only.

Layer 2 switches: These switches operate at the Data Link layer of the OSI model and are responsible for forwarding data between devices on the same network segment.

Layer 3 switches: These switches operate at the Network layer of the OSI model and can route data between different network segments. They are more advanced than Layer 2 switches and are often used in larger, more complex networks.

Routers – A router is a device like a switch that routes data packets based on their IP addresses. The router is mainly a Network Layer device. Routers normally connect LANs and WANs and have a dynamically updating routing table based on which they make decisions on routing the data packets. The router divides the broadcast domains of hosts connected through it.

Gateway – A gateway, as the name suggests, is a passage to connect two networks that may work upon different networking models. They work as messenger agents that take data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switches or routers. A gateway is also called a protocol converter.

NIC – NIC or network interface card is a network adapter that is used to connect the computer to the network. It is installed in the computer to establish a LAN. It has a unique id that is written on the chip, and it has a connector to connect the cable to it. The cable acts as an interface between the computer and the router or modem. NIC card is a layer 2 device which means that it works on both the physical and data link layers of the network model.

Multiplexing

Multiplexing is a technique used in telecommunications and computer networks to combine multiple signals into a single transmission channel. It allows multiple devices to share a common communication channel, which increases the efficiency and capacity of the network.

OSI(Open System Interconnection)	TCP/IP(Transmission Control Protocol / Internet Protocol)
1. OSI is a generic, protocol independent standard, acting as a communication gateway between the network and end user.	1. TCP/IP model is based on standard protocols around which the Internet has developed. It is a communication protocol, which allows connection of hosts over a network.
2. In OSI model the transport layer guarantees the delivery of packets.	2. In TCP/IP model the transport layer does not guarantees delivery of packets. Still the TCP/IP model is more reliable.
3. Follows horizontal approach.	3. Follows vertical approach.
4. OSI model has a separate Presentation layer and Session layer.	4. TCP/IP does not have a separate Presentation layer or Session layer.
5. Transport Layer is Connection Oriented.	5. Transport Layer is both Connection Oriented and Connection less.
6. Network Layer is both Connection Oriented and Connection less.	6. Network Layer is Connection less.
7. OSI is a reference model around which the networks are built. Generally it is used as a guidance tool.	7. TCP/IP model is, in a way implementation of the OSI model.
8. Network layer of OSI model provides both connection oriented and connectionless service.	8. The Network layer in TCP/IP model provides connectionless service.
9. OSI model has a problem of fitting the protocols into the model.	9. TCP/IP model does not fit any protocol
10. Protocols are hidden in OSI model and are easily replaced as the technology changes.	10. In TCP/IP replacing protocol is not easy.
11. OSI model defines services, interfaces and protocols very clearly and makes clear distinction between them. It is protocol independent.	11. In TCP/IP, services, interfaces and protocols are not clearly separated. It is also protocol dependent.
12. It has 7 layers	12. It has 4 layers